Portugal's Economic and Human Development: A Century of Transformation

Data Analysis and Integration

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Task 1

Staging database creation

1

```
CREATE DATABASE IF NOT EXISTS staging;
USE staging;

DROP TABLE IF EXISTS staging_maddison;
CREATE TABLE staging_maddison (
    country_iso3    CHAR(3) NOT NULL,
    country_name    VARCHAR(20) NOT NULL,
    indicator_code  VARCHAR(20) NOT NULL,
    indicator_name  VARCHAR(100)    NOT NULL,
    year            SMALLINT    NOT NULL,
    value           DOUBLE  NULL
);
```

2

```
DROP TABLE IF EXISTS staging_wdi;
CREATE TABLE staging_wdi (
    country_iso3    CHAR(3)  NOT NULL,
    country_name    VARCHAR(20) NOT NULL,
    indicator_code  VARCHAR(20) NOT NULL,
    indicator_name  VARCHAR(100) NOT NULL,
    year            SMALLINT NOT NULL,
    value            DOUBLE  NULL
);
```

3

```
DROP TABLE IF EXISTS staging_historical_events;
CREATE TABLE staging_historical_events (
    year            SMALLINT NOT NULL,
    country_iso3    CHAR(3) NOT NULL,
    country_name    VARCHAR(20) NOT NULL,
    event           VARCHAR(150) NOT NULL,
    category        VARCHAR(50) NOT NULL,
    economic_impact VARCHAR(50) NULL
);
```

Created a staging database with three source-specific tables and loaded the supplied CSV files to preserve the source structure and provide a consistent SQL environment for data profiling before transformation.

source	total_rows
maddison	4349
wdi	11371
historical_events	387

Task 1

Data profiling

Profiled dataset size, geographic scope and temporal coverage by source and indicator to establish the available coverage and identify limitations for subsequent analyses.

Dataset overview

```
SELECT 'maddison' AS source, COUNT(*) AS row_count,
       COUNT(DISTINCT country_iso3) AS num_entities,
       MIN(year) AS min_year, MAX(year) AS max_year
FROM staging_maddison
UNION ALL
SELECT 'wdi', COUNT(*), COUNT(DISTINCT country_iso3),
       MIN(year), MAX(year)
FROM staging_wdi
UNION ALL
SELECT 'historical_events', COUNT(*),
    COUNT(DISTINCT country_iso3), MIN(year), MAX(year)
FROM staging_historical_events;
```

source	row_count	num_entities	min_year	max_year
maddison	4349	15	1900	2022
wdi	11371	16	1960	2025
historical_events	387	16	1900	2025

Maddison covers 15 entities over 1900–2022. WDI and historical events cover 16 entities, over 1960–2025 and 1900–2025, respectively.

Madison Indicator Coverage

(The same indicator-level profiling was applied to WDI)

```
SELECT indicator_code, indicator_name,
       MIN(year) AS min_year, MAX(year) AS max_year,
       COUNT(*) AS row_count,
       COUNT(DISTINCT country_iso3) AS num_countries,
       COUNT(DISTINCT year) AS num_years_with_observations
FROM staging_maddison
GROUP BY indicator_code, indicator_name
ORDER BY min_year;
```

indicator_code	indicator_name	min_year	max_year	row_count	num_countries	num_years_with_observations
gdp	GDP (2011 international $)	1900	2022	1444	15	123
gdppc	GDP per capita (2011 international $)	1900	2022	1461	15	123
pop	Population (thousands)	1900	2022	1444	15	123

Each indicator spans 15 entities and 123 distinct years. A complete country–year panel would contain 1,845 observations per indicator; actual counts range from 1,444 to 1,461, revealing coverage gaps.

Task 1

Data profiling

Profiled year ranges and record counts by country, then examined Portugal across sources to identify country-specific coverage differences relevant to subsequent comparisons.

Coverage by country

```
SELECT source, country_iso3, country_name,
       MIN(year) AS min_year, MAX(year) AS max_year,
       COUNT(*) AS row_count
FROM (
    SELECT 'maddison' AS source, country_iso3, country_name, year
    FROM staging_maddison
    UNION ALL
    SELECT 'wdi' AS source, country_iso3, country_name, year
    FROM staging_wdi
    UNION ALL
    SELECT 'historical_events' AS source, country_iso3, country_name, year
    FROM staging_historical_events
) AS source_rows
GROUP BY source, country_iso3, country_name
ORDER BY source, country_name;
```

source	country_iso3	country_name	min_year	max_year	row_count
historical_events	BEL	Belgium	1900	2020	15
historical_events	CZE	Czechia	1918	2020	17
historical_events	EST	Estonia	1900	2020	21
historical_events	EUU	European Union	1957	2007	5
historical_events	FRA	France	1900	2020	19
historical_events	DEU	Germany	1900	2020	22

Country-level coverage differs from global coverage: Maddison observations for Czechia begin in 1970 and for Estonia in 1973.

Portugal across sources

```
SELECT 'maddison' as source, country_iso3, country_name,
    MIN(year) AS min_year, MAX(year) AS max_year
FROM staging_maddison sm
where sm.country_iso3 = 'PRT'
GROUP BY country_iso3, country_name
union all
SELECT 'wdi' as source, country_iso3, country_name,
    MIN(year) AS min_year, MAX(year) AS max_year
FROM staging_wdi
where country_iso3 = 'PRT'
GROUP BY country_iso3, country_name
union all
SELECT 'historical' as source, country_iso3, country_name,
    MIN(year) AS min_year, MAX(year) AS max_year
from staging_historical_events she
where she.country_iso3 = 'PRT'
GROUP BY country_iso3, country_name;
```

source	country_iso3	country_name	min_year	max_year
maddison	PRT	Portugal	1900	2022
wdi	PRT	Portugal	1960	2025
historical	PRT	Portugal	1900	2025

Portugal has a recorded span of 1900–2022 in Maddison and 1960–2025 in WDI. Historical events extend to 2025 and provide context for interpreting economic changes.

Task 1

Data profiling

Checked duplicate keys and NULL values across the three sources to identify repeated observations and missing fields before transformation.

Duplicate keys

```sql
SELECT country_iso3, indicator_code, year, COUNT(*) AS n
FROM staging_wdi
GROUP BY country_iso3, indicator_code, year
HAVING COUNT(*) > 1;
```

```sql
SELECT country_iso3, indicator_code, year, COUNT(*) AS n
FROM staging_maddison
GROUP BY country_iso3, indicator_code, year
HAVING COUNT(*) > 1;
```

```sql
SELECT year, country_iso3, event, COUNT(*) AS n
FROM staging_historical_events
GROUP BY year, country_iso3, event
HAVING COUNT(*) > 1;
```

No repeated keys were detected. Indicator data were checked by country–indicator–year, and events by year–country–event.

NULL checks

```sql
SELECT COUNT(*) AS null_values FROM staging_wdi WHERE value IS NULL;
SELECT COUNT(*) AS null_values FROM staging_maddison WHERE value IS NULL;
```

```sql
SELECT
    COUNT(*) - COUNT(year) AS null_years,
    COUNT(*) - COUNT(country_iso3) AS null_country_codes,
    COUNT(*) - COUNT(country_name) AS null_country_names,
    COUNT(*) - COUNT(event) AS null_events,
    COUNT(*) - COUNT(category) AS null_categories,
    COUNT(*) - COUNT(economic_impact) AS null_economic_impacts
FROM staging_historical_events;
```

wdi, maddison

null_values
0

Historical events — NULL values by column

null_years	null_country_codes	null_country_names	null_events	null_categories	null_economic_impacts
0	0	0	0	0	0

No NULL values were detected in the checked fields. However, absent observations and empty strings are not detected by these queries.

Task 1

Historical events data dictionary

Historical events data dictionary

Source file: `project_data_sources_2627/historical_events.csv`
Staging table: `staging_historical_events`
Rows: 387 · **Entities:** 16 (15 countries and the European Union aggregate) · **Observed year span:** 1900–2025 · **Distinct years with events:** 110

Columns

Column	Staging type	Null?	Description / units
`year`	`SMALLINT`	No	Year assigned to the event; no measurement unit.
`country_iso3`	`CHAR(3)`	No	Three-letter country or aggregate identifier.
`country_name`	`VARCHAR(20)`	No	Country or aggregate name.
`event`	`VARCHAR(150)`	No	Narrative description of the historical event.
`category`	`VARCHAR(50)`	No	Categorical event classification.
`economic_impact`	`VARCHAR(50)`	Yes	Qualitative text describing an economic impact; not a numeric measure.

Coverage

Geographic coverage: Belgium (BEL), Czechia (CZE), Estonia (EST), France (FRA), Germany (DEU), Greece (GRC), Hungary (HUN), Ireland (IRL), Italy (ITA), Latvia (LVA), Lithuania (LTU), Poland (POL), Portugal (PRT), Romania (ROU), Spain (ESP), and the European Union aggregate (EUU).

Missingness and notable patterns

Complete dictionary is available in *data_dictionary_events.pdf*

Historical context

The 387 events cover 15 countries and the EU aggregate over 1900–2025, providing qualitative context for economic and social turning points.

Curated coverage

Events occur in 110 distinct years. Years without entries do not necessarily indicate missing data, because the dataset is a curated event collection rather than an annual series.

Formatting and integrity

An unquoted comma in the Ireland 1990 description was corrected. The dictionary reports six fields per record, no empty fields and no duplicate year–country–event combinations.

Interpretation and integration

Categories and economic impacts are qualitative labels, not measured effects. Multiple events may occur in one country–year, requiring care when joining them to numeric indicators.

Task 1

Maddison indicators data dictionary

Maddison indicators data dictionary

Source file: `project_data_sources_2627/maddison_indicators.csv`
Staging table: `staging_maddison`
Rows: 4,349 · **Entities:** 15 countries · **Observed year span:** 1900–2022

Columns

Column	Staging type	Null?	Description / units
`country_iso3`	`CHAR(3)`	No	Three-letter country identifier.
`country_name`	`VARCHAR(20)`	No	Country name.
`indicator_code`	`VARCHAR(20)`	No	Measure code: `gdp`, `gdppc`, or `pop`.
`indicator_name`	`VARCHAR(100)`	No	Human-readable measure name.
`year`	`SMALLINT`	No	Observation year.
`value`	`DOUBLE`	Yes	Numeric measure; units depend on the indicator below.

Measures and coverage

Code	Measure and unit	Rows	Countries	Observed year span	Absent country-year rows*
`gdp`	GDP (2011 international dollars)	1,444	15	1900–2022	401
`gdppc`	GDP per capita (2011 international dollars)	1,461	15	1900–2022	384
`pop`	Population (thousands)	1,444	15	1900–2022	401

Geographic coverage: Belgium (BEL), Czechia (CZE), Estonia (EST), France (FRA), Germany (DEU), Greece

Complete dictionary is available in *data_dictionary_maddison.pdf*

Long-run analysis

The dataset supports long-run comparisons of GDP, GDP per capita and population across 15 countries from 1900 to 2022. No EU aggregate is available.

Uneven country coverage

Coverage begins later for Ireland (1913), Czechia (1970) and the Baltic countries (1973), restricting early comparisons with Portugal.

Missing combinations

No explicit missing values or duplicate observation keys were identified. However, GDP per capita has 384 absent country–year combinations relative to a complete 15-country annual grid.

Measurement implications

GDP measures use 2011 international dollars, while population is measured in thousands. Units must be preserved during integration and interpretation.

Task 1

WDI data dictionary

World Development Indicators (WDI) data dictionary

Source file: `project_data_sources_2627/wdi_indicators.csv` **Staging table:** `staging_wdi` **Rows:** 11,371 · **Entities:** 16 (15 countries and the European Union aggregate) · **Observed year span:** 1960–2025

Columns

Column	Staging type	Null?	Description / units
`country_iso3`	`CHAR(3)`	No	Three-letter country or aggregate identifier.
`country_name`	`VARCHAR(20)`	No	Country or aggregate name.
`indicator_code`	`VARCHAR(20)`	No	World Development Indicators measure code.
`indicator_name`	`VARCHAR(100)`	No	Human-readable measure name.
`year`	`SMALLINT`	No	Observation year.
`value`	`DOUBLE`	Yes	Numeric measure; units depend on the indicator below.

Measures and coverage

Code	Measure and unit	Rows	Entities	Observed year span	Absent entity-year rows*
`NE.EXP.GNFS.ZS`	Exports of goods and services (% of GDP)	757	16	1960–2025	299
`NE.IMP.GNFS.ZS`	Imports of goods and services (% of GDP)	757	16	1960–2025	299
`NV.AGR.TOTL.CD`	Agriculture, forestry, and fishing value added (current US$)	547	16	1960–2025	509

Complete dictionary is available in *data_dictionary_maddison.pdf*

Complementary indicators

WDI provides 17 economic, social and sectoral indicators for 16 entities, including the EU aggregate, complementing Maddison's long-run measures.

Indicator-specific periods

Employment begins in 1991, GNI per capita in 1962 and GDP growth in 1961. Life expectancy ends in 2024, so the overall 1960–2025 span does not apply to every measure.

Different levels of completeness

Employment, life expectancy and urbanization have complete annual coverage within their periods. Other measures contain gaps, including 509 absent entity–year combinations for agricultural value added.

Comparability

Percentages, current US dollars and years represent different quantities. Analyses must use compatible units and verify coverage for the selected countries and indicators.

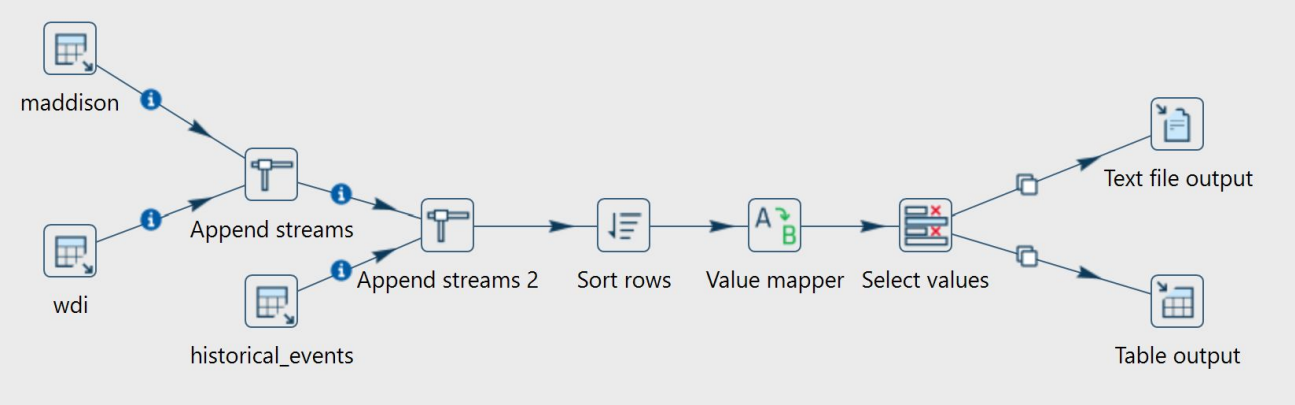
Task 1

Discussion and main findings

- No explicit missing values or duplicate keys were identified, but some country–indicator–year combinations are absent.
- Maddison GDP per capita has **384 absent country–year combinations** relative to a complete annual grid of 15 countries for 1900–2022.
- Maddison coverage starts in **1913 for Ireland, 1970 for Czechia and 1973 for the Baltic countries**, limiting comparisons before those dates.
- WDI employment indicators begin in **1991**, while life expectancy ends in **2024**; comparisons must use the periods available for each indicator.
- The **EU aggregate** appears only in WDI and historical events, so Maddison measures are unavailable for this entity.
- The **387 historical events cover 110 distinct years**. Years without recorded events should not automatically be treated as missing data, since this is a curated event list.

Task 2

Building the unified country mapping



Apache Hop pipeline: extraction, unification, group assignment and output to CSV and staging.

```sql
SELECT DISTINCT country_iso3, country_name
FROM staging_maddison;
```

Append Streams: combine the country lists from all three sources.
Sort rows: retain one record per ISO3 code.
Value mapper: Assign the required country_group.
Select values: rename country_iso3 to iso3_code and send the same records to both outputs.

Distinct country codes and names were extracted from each source. The same query was applied to staging_wdi and staging_historical_events. Only country identifiers and names are needed to build the mapping; repeated indicator and event records are excluded.

Aligned ISO3 codes simplify reconciliation, but a unified mapping is still required to assign country groups and include the EU aggregate, which is present only in WDI and historical events.

Task 2

Mapping outputs and validation

iso3_code	country_name	country_group
BEL	Belgium	Western Europe
CZE	Czechia	Central/Eastern Europe
DEU	Germany	Western Europe
ESP	Spain	Western Europe
EST	Estonia	Post-Soviet
EUU	European Union	Benchmark
FRA	France	Western Europe
GRC	Greece	Western Europe
HUN	Hungary	Central/Eastern Europe
IRL	Ireland	Western Europe
ITA	Italy	Western Europe
LTU	Lithuania	Post-Soviet
LVA	Latvia	Post-Soviet
POL	Poland	Central/Eastern Europe
PRT	Portugal	Focus
ROU	Romania	Central/Eastern Europe

country_mapping.csv

The EU aggregate is retained as a benchmark, but should not be added to individual country values, as this would double-count overlapping coverage.

```
SELECT country_group,
       COUNT(*) AS num_entities
FROM staging_country_mapping
GROUP BY country_group
ORDER BY country_group;
```

country_group	num_entities
Benchmark	1
Central/Eastern Europe	4
Focus	1
Post-Soviet	3
Western Europe	7

Validation: the mapping covers all five required groups, with 16 entities in total.

Unified coverage

The unified mapping contains 16 entities: 15 countries and the EU aggregate. Combining the three sources retains the EU, which is absent from Maddison.

Reconciliation approach

Since ISO3 codes and country names are already aligned, reconciliation focuses on combining and deduplicating the source lists, retaining one record per ISO3 code.

Country-group assignment

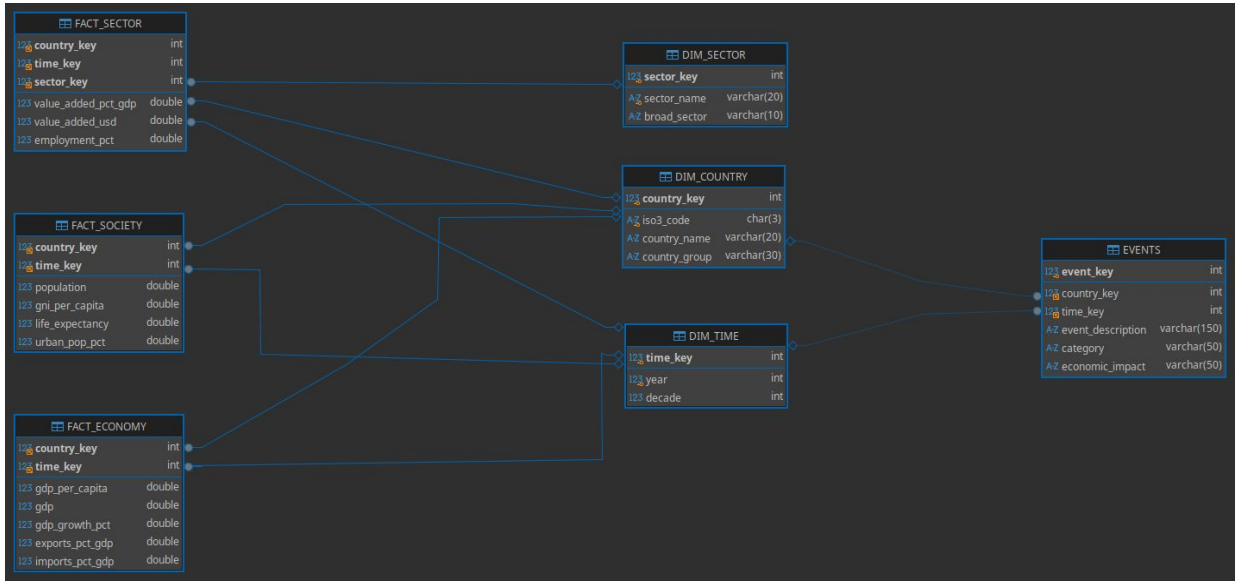
The five project-defined groups support comparisons between Portugal and its comparison countries. The EU is classified as Benchmark and interpreted as an aggregate rather than an individual country.

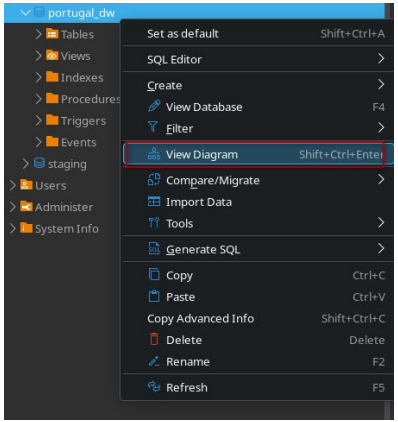
Output consistency

The same mapping is exported to CSV and loaded into staging. Validation confirms 16 records and 16 distinct ISO3 codes in both outputs, with matching names and groups.

Task 3

Data warehouse schema



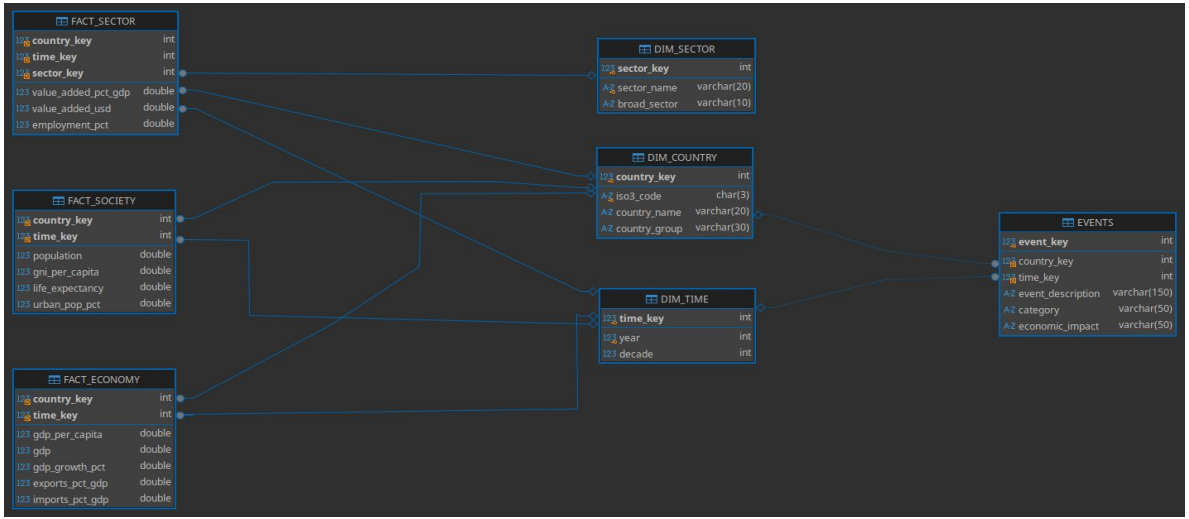


The ER diagram was generated using DBeaver's 'View Diagram' feature. Connector placement reflects the automatic layout; the SQL foreign-key definitions specify the exact column relationships.

DIM_COUNTRY and DIM_TIME support all three fact tables; DIM_SECTOR applies only to sector-level measures.
All dimensions use integer surrogate keys, independent of source identifiers.
EVENTS links countries and years, allowing multiple events per country–year without changing the fact-table grains.

Task 3

Data warehouse schema



Fact Table	Grain	Main Measures
FACT_ECONOMY	One row per country–year	GDP, GDP per capita, growth, exports and imports
FACT_SOCIETY	One row per country–year	Population, GNI per capita, life expectancy and urbanization
FACT_SECTOR	One row per country–year-sector	Value added and employment by sector

Task 3

SQL implementation (1)

Separate warehouse database

```
DROP DATABASE IF EXISTS portugal_dw;
CREATE DATABASE portugal_dw;
USE portugal_dw;
```

A separate database stores the dimensional model, keeping the warehouse structure independent from staging.

Created Schema

```
SHOW TABLES FROM portugal_dw;
```

Tables_in_portugal_dw
dim_country
dim_sector
dim_time
events
fact_economy
fact_sector
fact_society

SHOW TABLES confirms the creation of all seven expected tables: three dimensions, three fact tables and EVENTS. This verifies the schema structure; data loading is performed in the next task.

Complete implementation is provided in create_dw.sql

Task 3

Keys and referential integrity

Dimension Keys

```
CREATE TABLE DIM_COUNTRY (
    country_key      INT AUTO_INCREMENT,
    iso3_code        CHAR(3),
    country_name     VARCHAR(20),
    country_group    VARCHAR(30),
    PRIMARY KEY (country_key),
    UNIQUE (iso3_code)
);
```

Each dimension uses an automatically generated integer surrogate key as its primary key. Natural identifiers are retained for lookup and protected by UNIQUE constraints.
DIM_COUNTRY uses an automatically generated surrogate key to identify each record. ISO3 codes are retained for source matching, with a UNIQUE constraint preventing repeated non-NULL codes.

Fact References and Grain

```
UNIQUE (country_key, time_key, sector_key),
FOREIGN KEY (country_key) REFERENCES DIM_COUNTRY (country_key),
FOREIGN KEY (time_key)    REFERENCES DIM_TIME (time_key),
FOREIGN KEY (sector_key)  REFERENCES DIM_SECTOR (sector_key)
```

Foreign Keys:
Non-NULL country, time and sector keys must match existing dimension records, preventing references to nonexistent countries, years or sectors.

UNIQUE constraint:
When all three keys are populated, the UNIQUE constraint prevents duplicate country–year–sector combinations.

Complete implementation is provided in create_dw.sql

Task 3

Design Decisions and Discussion

Why separate fact tables?

Economy and Society share a country–year grain but describe different subject areas.

Sector has a finer country–year–sector grain. Keeping it separate avoids repeating national GDP or population for each sector.

DIM_SECTOR is therefore used only where a sector breakdown exists.

Why surrogate keys?

Surrogate keys separate warehouse identifiers from source codes and descriptions, while natural identifiers remain available for lookup.

UNIQUE constraints define the intended fact-table grains.

Foreign keys ensure that non-NULL country, time and sector references correspond to existing dimension records.

Why an EVENTS bridge?

Linking events to both country and year provides more precise context than a year-only reference table.

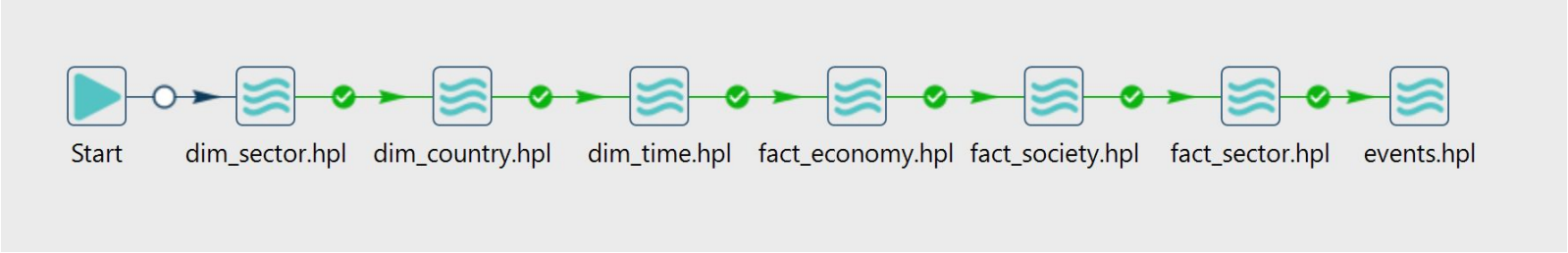
A separate EVENTS table supports multiple events per country–year without adding event-level detail to the indicator facts.

No DIM_EVENT was created because events are used as descriptive context rather than as a dimension for analysing measures.

Multiple events can match one fact row. Retrieve events separately or aggregate them by country–year before joining, to avoid duplicating measures.

Task 4

ETL workflow and execution order



dw_workflow.hwf orchestrates the seven ETL pipelines in a sequential execution

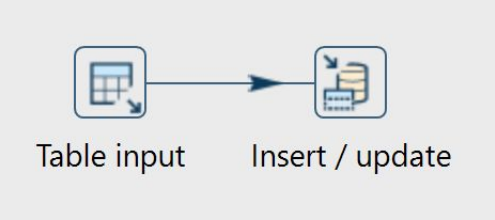
Loading dependencies → Dimensions are loaded before facts and events so that surrogate keys are available for lookup. This order reduces the risk of unresolved references during loading.

Execution control and limitations → Sequential execution and success-conditioned hops prevent downstream pipelines from running after a failure. However, earlier loads may already have been committed, so a failed workflow does not automatically roll back the entire warehouse.

All seven pipelines completed successfully; the resulting table counts are validated in the following slides.

Task 4

DIM_COUNTRY: Loading the country dimension



Loads the unified country mapping into DIM_COUNTRY

Table input

Transform name: Table input

Connection: staging

Load SQL from file:

SQL — Get SQL select

```
SELECT iso3_code, country_name, country_group
FROM staging_country_mapping
```

Insert / update

Transform name: Insert / update

General | Lookup keys | Fields

Connection: staging

Target schema: portugal_dw

Target table: DIM_COUNTRY

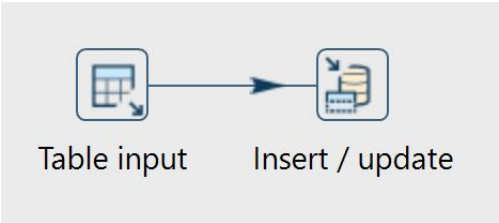
Commit size: 100

Don't perform any updates:

#	Table field	Stream field	Update
1	iso3_code	iso3_code	Y
2	country_name	country_name	Y
3	country_group	country_group	Y

Task 4

DIM_COUNTRY: Loading the country dimension



Loads the unified country mapping into DIM_COUNTRY

```
SELECT COUNT(*) AS total_rows,
       COUNT(DISTINCT iso3_code) AS distinct_codes
FROM portugal_dw.DIM_COUNTRY;
```

total_rows	distinct_codes
16	16

The loaded dimension contains 16 records and 16 distinct ISO3 codes, confirming one record per entity.

Why Insert / update?

Insert / update supports repeated loads by matching existing countries before writing. Table output alone inserts rows and would require a separate strategy to handle existing records.

Preserving surrogate keys

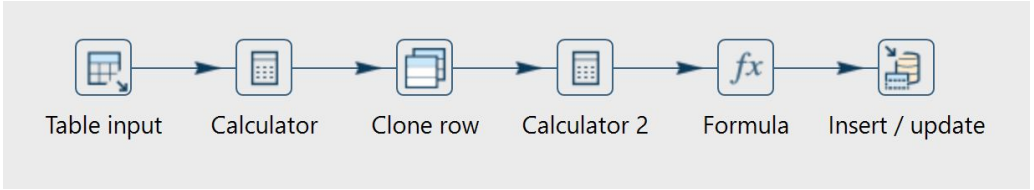
Updating records in place preserves existing country_key values. Deleting and reloading dimension records could regenerate keys and disrupt references from facts and events.

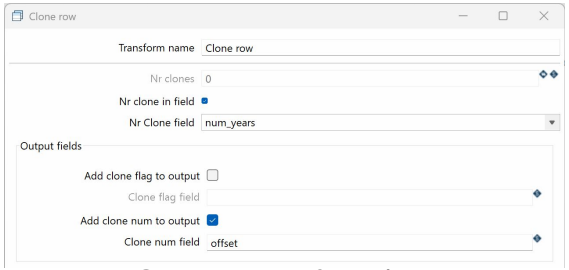
Current-state limitation

Changes overwrite existing names and groups, so attribute history is not retained. Records removed from the source mapping are not automatically deleted.

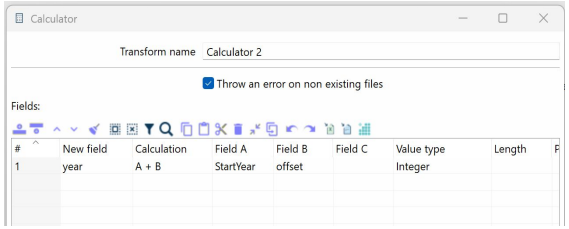
Task 4

DIM_TIME: Year and decade generation

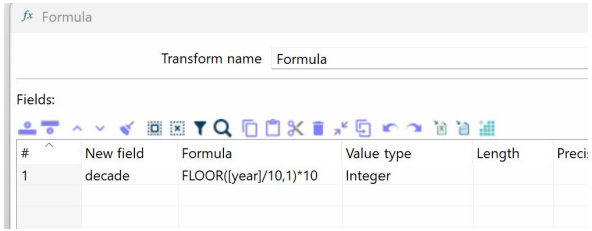




Creates one row for each year



Calculates the calendar year



Assigns each year to its decade

```
SELECT COUNT(*) AS total_rows,
       COUNT(DISTINCT year) AS distinct_years,
       MIN(year) AS first_year,
       MAX(year) AS last_year
FROM portugal_dw.DIM_TIME;
```

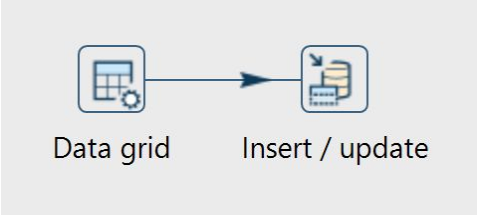
total_rows	distinct_years	first_year	last_year
126	126	1900	2025

DIM_TIME contains 126 unique years, covering the complete 1900–2025 range across all three sources.

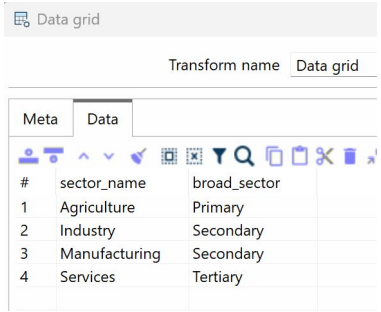
The continuous calendar supports year and decade comparisons. A year's presence in DIM_TIME does not imply that indicators or events are available for that year.

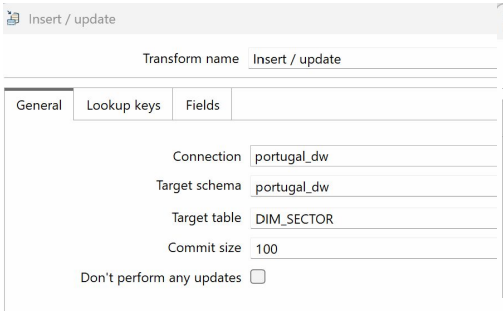
Task 4

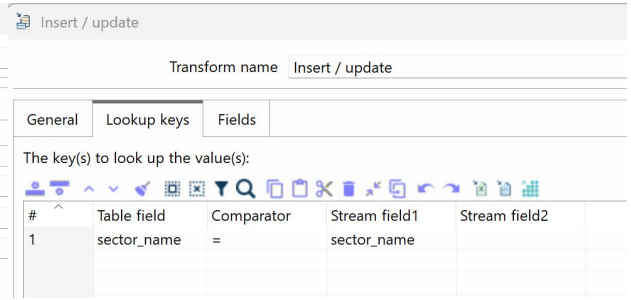
DIM_SECTOR: Sector hierarchy



Loads the predefined sectors and their broad-sector groups into DIM_SECTOR







Four sectors are assigned to three broad-sector groups, establishing the Broad Sector → Sector hierarchy

Matching by sector_name refreshes existing records while preserving their surrogate sector_key

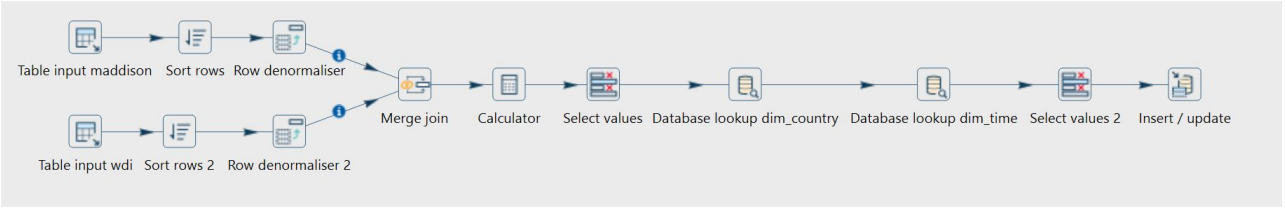
```
SELECT COUNT(*) AS total_rows,
       COUNT(DISTINCT sector_name) AS distinct_sectors,
       COUNT(DISTINCT broad_sector) AS broad_groups
FROM portugal_dw.DIM_SECTOR;
```

total_rows	distinct_sectors	broad_groups
4	4	3

DIM_SECTOR contains four unique sectors across three broad groups, providing consistent categories for sector-key lookups and hierarchical analysis. Manufacturing is part of Industry. Their measures should not be added together, as this would double-count overlapping activity.

Task 4

FACT_ECONOMY: Economic indicators



For maddison:

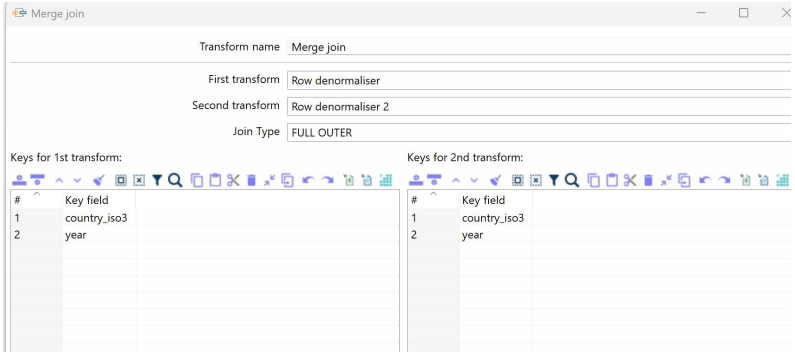
#	Target fieldname	Value fieldname	Key value	Type
1	gdp	value	gdp	Number
2	gdp_per_capita	value	gdppc	Number

For wdi:

Target fields:

#	Target fieldname	Value fieldname	Key value	Type
1	gdp_growth_pct	value	NY.GDP.MKTP.KD.ZG	Number
2	exports_pct_gdp	value	NE.EXP.GNFS.ZS	Number
3	imports_pct_gdp	value	NE.IMP.GNFS.ZS	Number

Row denormalisers convert indicator rows into measure columns, producing one record per country–year in each source stream

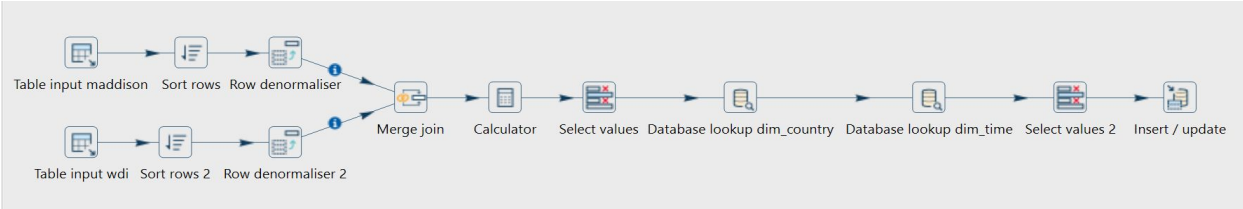


The FULL OUTER join retains country–year observations available in either source, preserving their different temporal and geographic coverage.

Measures unavailable in one source remain NULL. These represent coverage gaps and should not be interpreted as zero.

Task 4

FACT_SOCIETY: Social and demographic indicators



For maddison:

Target fields:

Get lool

#	Target fieldname	Value fieldname	Key value	Type	Forn
1	population	value	pop	Number	

The fields that make up the grouping:

#	Group field
1	country_iso3
2	year
3	

For wdi:

Target fields:

#	Target fieldname	Value fieldname	Key value	Type
1	gni_per_capita	value	NY.GNP.PCAP.CD	Number
2	life_expectancy	value	SP.DYN.LE00.IN	Number
3	urban_pop_pct	value	SP.URB.TOTL.IN.ZS	Number

Merge join

Transform name: Merge join

First transform: Row denormaliser

Second transform: Row denormaliser 2

Join Type: FULL OUTER

Keys for 1st transform:

#	Key field
1	country_iso3
2	year

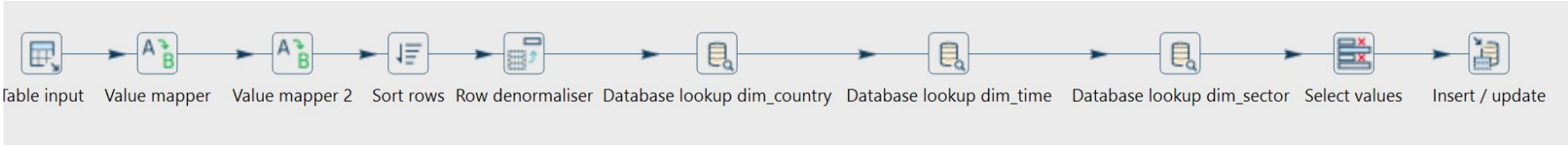
Keys for 2nd transform:

#	Key field
1	country_iso3
2	year

The FULL OUTER join preserves observations available in either source, combining long-run population coverage with more recent social indicators.
Population remains in the Maddison source unit of thousands. Missing measures reflect source coverage gaps and should not be interpreted as zero.

Task 4

FACT_SECTOR: Sector mapping and transformation



Field values:

#	Source value	Target value	Empty string ec
1	NV.AGR.TOTL.ZS	Agriculture	Y
2	NV.AGR.TOTL.CD	Agriculture	Y
3	NV.IND.TOTL.ZS	Industry	Y
4	NV.IND.TOTL.CD	Industry	Y
5	NV.IND.MANF.ZS	Manufacturing	Y
6	NV.IND.MANF.CD	Manufacturing	Y
7	NV.SRV.TOTL.ZS	Services	Y
8	NV.SRV.TOTL.CD	Services	Y
9	SL.AGR.EMPL.ZS	Agriculture	Y
10	SL.IND.EMPL.ZS	Industry	Y
11	SL.SRV.EMPL.ZS	Services	Y

Value Mapper 1 - sector_name

Field values:

#	Source value	Target value	Empty string equ
1	NV.AGR.TOTL.ZS	value_added_pct_gdp	Y
2	NV.AGR.TOTL.CD	value_added_usd	Y
3	NV.IND.TOTL.ZS	value_added_pct_gdp	Y
4	NV.IND.TOTL.CD	value_added_usd	Y
5	NV.IND.MANF.ZS	value_added_pct_gdp	Y
6	NV.IND.MANF.CD	value_added_usd	Y
7	NV.SRV.TOTL.ZS	value_added_pct_gdp	Y
8	NV.SRV.TOTL.CD	value_added_usd	Y
9	SL.AGR.EMPL.ZS	employment_pct	Y
10	SL.IND.EMPL.ZS	employment_pct	Y
11	SL.SRV.EMPL.ZS	employment_pct	Y

Value Mapper 2 - measure_type

The fields that make up the grouping:

#	Group field
1	country_iso3
2	year
3	sector_name

Target fields:

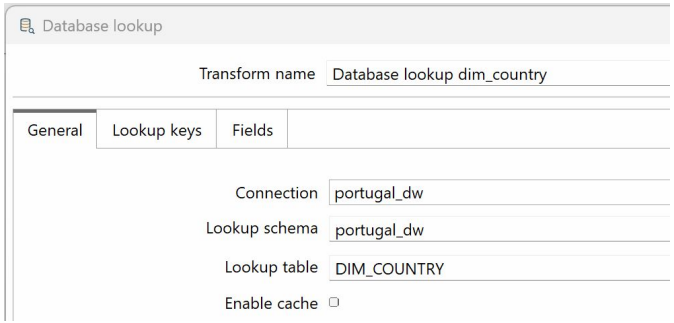
#	Target fieldname	Value fieldname	Key value	Type
1	value_added_pct_gdp	value	value_added_pct_gdp	Number
2	value_added_usd	value	value_added_usd	Number
3	employment_pct	value	employment_pct	Number

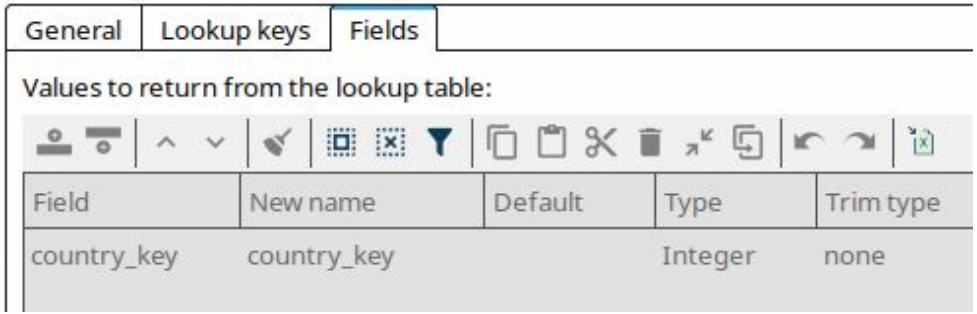
Row dernomaliser

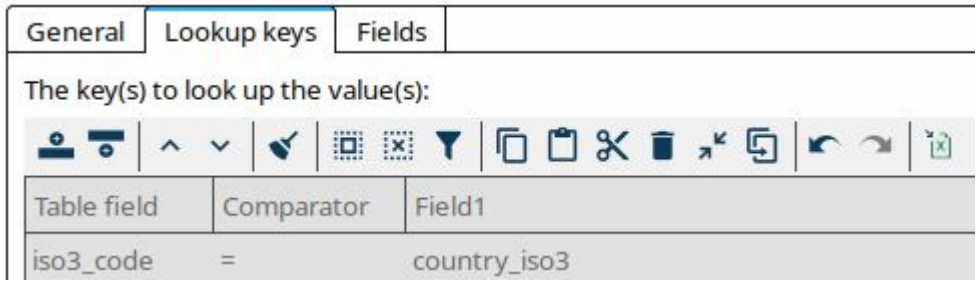
Separating sector identification from measure type aligns value added and employment within the same sector record. Manufacturing has no employment indicator in the selected source data, so employment_pct remains NULL for this sector.

Task 4

FACT_SECTOR: Key lookups and loading results

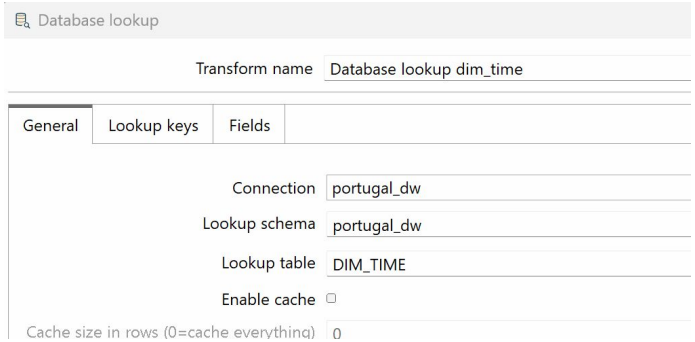


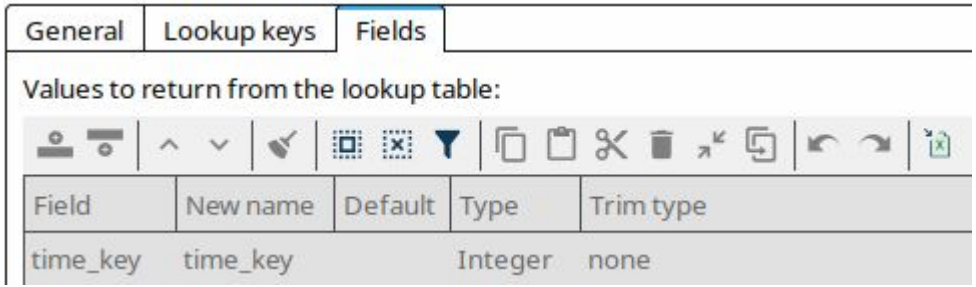


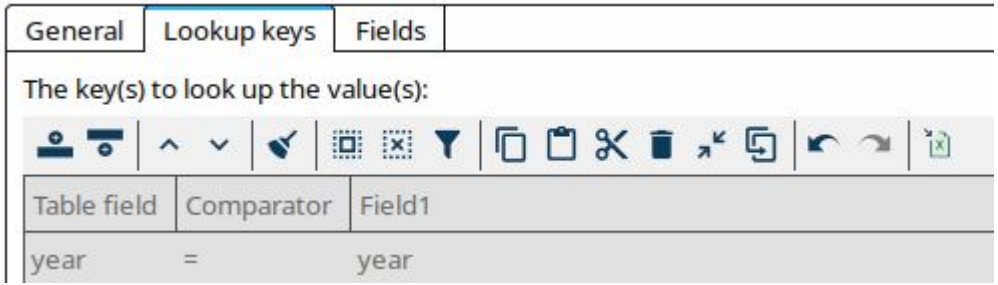


Task 4

FACT_SECTOR: Key lookups and loading results

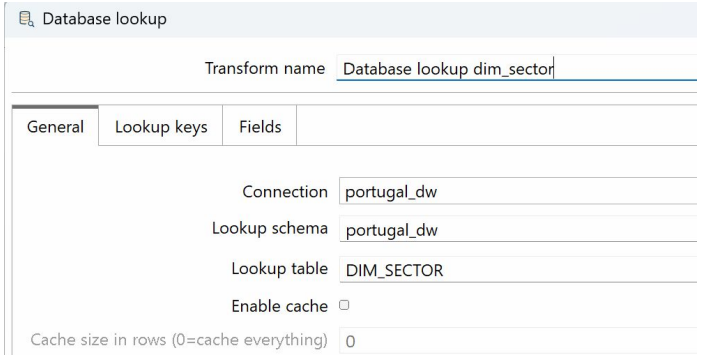


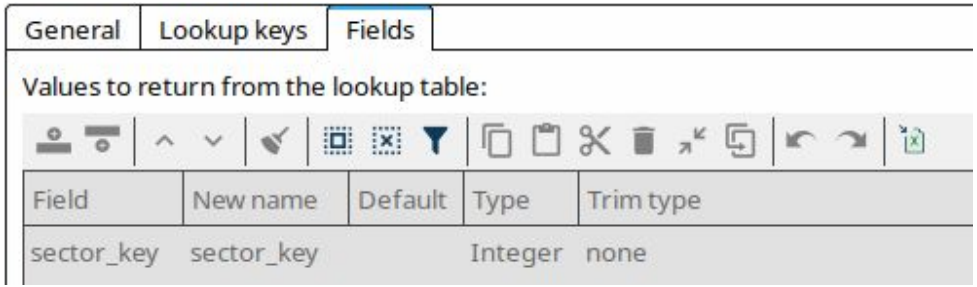


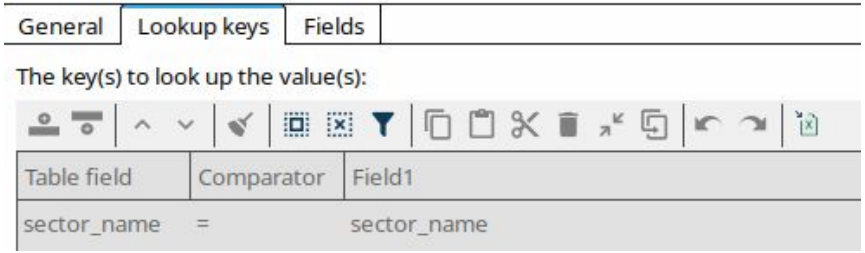


Task 4

FACT_SECTOR: Key lookups and loading results

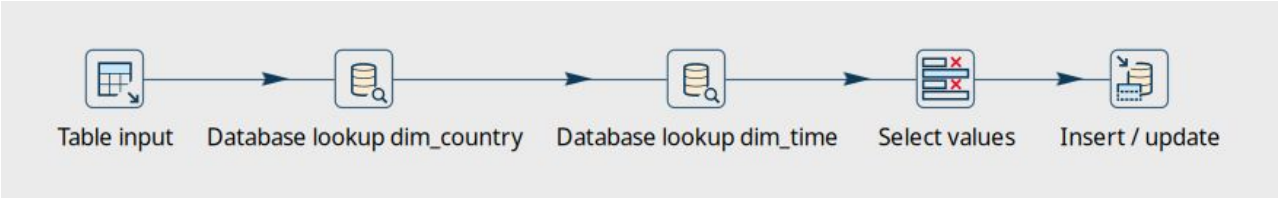






Task 4

EVENTS: Historical context loading



Transform name: Table input

Connection: staging

SQL

```
SELECT country_iso3, year, event, category, economic_impact
FROM staging_historical_events
```

Table Input

General | Lookup keys | Fields

Values to return from the lookup table:

Field	New name	Default	Type	Trim type
country_key	country_key		Integer	none

Transform name: Database lookup dim_country

General | Lookup keys | Fields

The key(s) to look up the value(s):

Table field	Comparator	Field1
iso3_code	=	country_iso3

Lookup1
dim_country

Transform name: Database lookup dim_country

Connection: portugal_dw

Lookup schema: portugal_dw

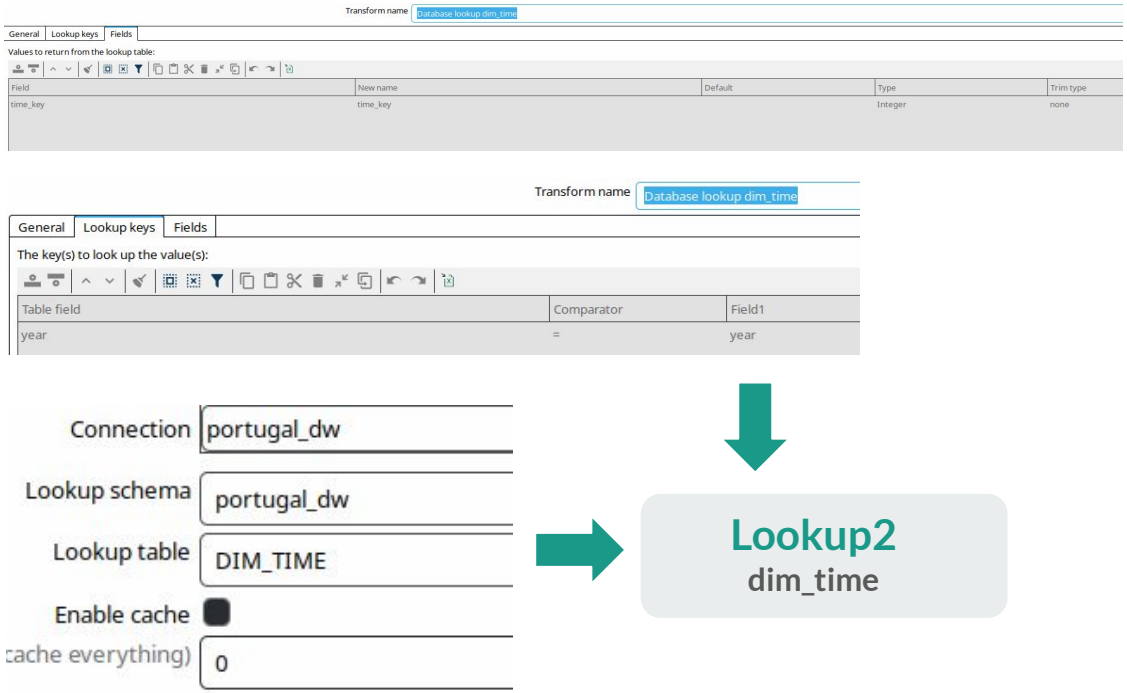
Lookup table: DIM_COUNTRY

Enable cache

ws (0=cache everything): 0

Task 4

EVENTS: Historical context loading



Each lookup resolves a natural key (iso3 code / year) into its surrogate key, linking events to the same conformed dimensions used by the fact tables.

Task 4

EVENTS: Historical context loading

Select values

Select & Alter | Remove | Meta-data

Fields :

#	Fieldname	Rename to	Length	Precision
1	country_key			
2	time_key			
3	event	event_description		
4	category			
5	economic_impact			

Insert/update

Transform name: Insert / update
Connection: portugal_dw
Target schema: portugal_dw
Target table: EVENTS
Commit size: 100
Don't perform any updates

The key(s) to look up the value(s):

#	Table field	Comparator	Stream field1	Stream field2
1	country_key	=	country_key	
2	time_key	=	time_key	
3	event_description	=	event_description	

Update fields:

#	Table field	Stream field	Update
1	country_key	country_key	Y
2	time_key	time_key	Y
3	event_description	event_description	Y
4	category	category	Y
5	economic_impact	economic_impact	Y

Task 4

EVENTS: Historical context loading

```
SELECT COUNT(*) AS total_rows,
       COUNT(*) - COUNT(country_key) AS null_country_keys,
       COUNT(*) - COUNT(time_key) AS null_time_keys
FROM portugal_dw.EVENTS;
```

total_rows	null_country_keys	null_time_keys
387	0	0

Validation query: counts loaded events and checks for missing country and time keys.
387 events loaded; no missing country or time keys.

Validation results

The loaded table contains 387 events, with no missing country or time keys. This confirms that all loaded records have dimensional references; the row-count check alone does not establish uniqueness or complete source-to-target equivalence.

Loading strategy

Matching records by country, year and event description supports repeated loads without reinserting exact matches. However, changes to an event description may create a new record rather than update the existing event.

Analytical use and limitations

Historical events help interpret observed turning points, but their qualitative impact labels do not demonstrate causality. Since several events may share a country and year, direct joins to fact tables can repeat measures and distort totals.

Task 4

ETL validation and discussion

```
SELECT 'DIM_COUNTRY' AS table_name, COUNT(*) AS row_count
FROM portugal_dw.DIM_COUNTRY
UNION ALL
SELECT 'DIM_TIME', COUNT(*) FROM portugal_dw.DIM_TIME
UNION ALL
SELECT 'DIM_SECTOR', COUNT(*) FROM portugal_dw.DIM_SECTOR
UNION ALL
SELECT 'FACT_ECONOMY', COUNT(*) FROM portugal_dw.FACT_ECONOMY
UNION ALL
SELECT 'FACT_SOCIETY', COUNT(*) FROM portugal_dw.FACT_SOCIETY
UNION ALL
SELECT 'FACT_SECTOR', COUNT(*) FROM portugal_dw.FACT_SECTOR
UNION ALL
SELECT 'EVENTS', COUNT(*) FROM portugal_dw.EVENTS;
```

Row counts confirm that all seven warehouse tables are populated. Different fact-table counts reflect their distinct grains and source coverage.

table_name	row_count
DIM_COUNTRY	16
DIM_TIME	126
DIM_SECTOR	4
FACT_ECONOMY	1571
FACT_SOCIETY	1622
FACT_SECTOR	2329
EVENTS	387

Grain transformation

Pivoting combines multiple indicator rows into country–year or country–year–sector records. Fact-table counts therefore differ from staging counts.

Coverage preservation

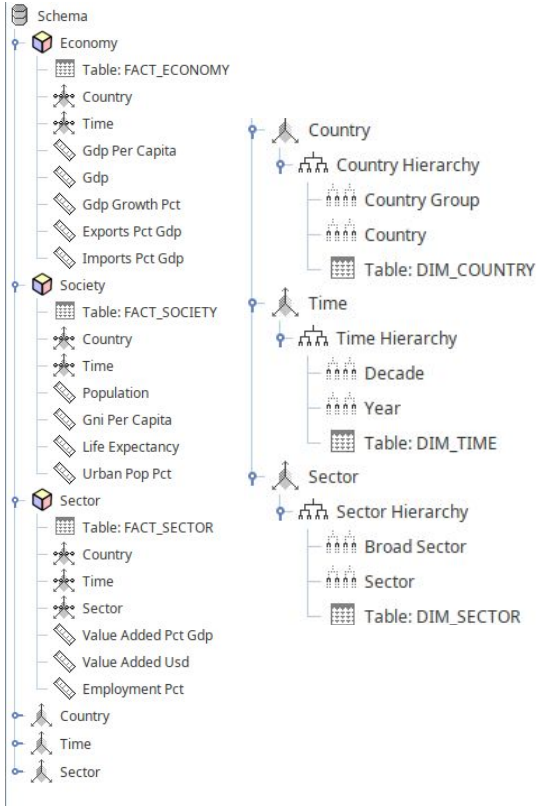
FULL OUTER joins preserve observations from either source. Missing measures remain NULL, reflecting uneven coverage rather than zero values.

Repeatable loading

Insert/update matches existing records using dimension identifiers or fact-grain keys, allowing attributes and measures to be refreshed without duplicating successfully matched records.
Validation confirmed unique fact-grain combinations and no missing dimension keys. These checks support structural consistency but do not establish complete measure coverage.

Task 5

Cube definition: schema overview



Cube	Fact table	Main measures
Economy	FACT_ECONOMY	GDP, GDP per capita, growth and trade
Society	FACT_SOCIETY	Population, GNI per capita, life expectancy and urbanization
Sector	FACT_SECTOR	Value added and employment

Subject-area separation

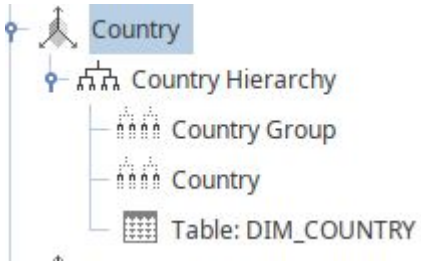
Three cubes expose measures from the three fact tables while preserving their respective grains. Economy and Society support country–year analysis; Sector adds sector-level detail.

Consistent navigation

Shared Country and Time dimensions provide consistent filters and hierarchy levels across the cubes.

Task 5

Country hierarchy



Country Group → Country enables regional comparisons and country-level drill-down
Shared Country Dimension used across all three cubes

Dimension table and key

Table for 'Country Hierarchy' Hierarchy

Attribute	Value
schema	
name	DIM_COUNTRY
alias	

Hierarchy for 'Country' Dimension

Attribute	Value
name	Country Hierarchy
description	
hasAll	☑
allMemberName	All Countries
allMemberCaption	
allLevelName	
defaultMember	
memberReaderClass	
primaryKeyTable	
primaryKey	country_key
caption	
visible	☑

Hierarchy levels

Level for 'Country Hierarchy' Hierarchy

Attribute	Value
name	Country
description	
table	
column	country_name
nameColumn	
parentColumn	
nullParentValue	
ordinalColumn	
type	String
internalType	
uniqueMembers	☐
levelType	Regular
hideMemberIf	Never
approxRowCount	
caption	
captionColumn	
formatter	
visible	☑

Level for 'Country Hierarchy' Hierarchy

Attribute	Value
name	Country Group
description	
table	
column	country_group
nameColumn	
parentColumn	
nullParentValue	
ordinalColumn	
type	String
internalType	
uniqueMembers	☐
levelType	Regular
hideMemberIf	Never
approxRowCount	
caption	
captionColumn	
formatter	
visible	☑

Comparison design

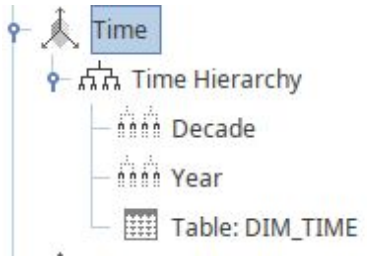
The hierarchy separates Portugal as the Focus country and organizes comparison countries into regional groups. This supports the required Western and Eastern European comparisons while allowing drill-down to identify differences hidden by group averages.

Interpretation of results

Regional results depend on the selected measure: AVG gives an unweighted average of available observations, not a population-weighted regional indicator. The EU Benchmark is a separate aggregate and should not be combined with member-country totals

Task 5

Time hierarchy



Decade → Year supports long-run comparisons and annual drill-down
Shared TimeDimension used across all three cubes

Dimension table and key

Table for 'Time Hierarchy' Hierarchy

Attribute	Value
schema	
name	DIM_TIME
alias	

Hierarchy for 'Time' Dimension

Attribute	Value
name	Time Hierarchy
description	
hasAll	☑
allMemberName	All Times
allMemberCaption	
allLevelName	
defaultMember	
memberReaderClass	
primaryKeyTable	
primaryKey	time_key
caption	
visible	☑

Hierarchy levels

Level for 'Time Hierarchy' Hierarchy

Attribute	Value
name	Decade
description	
table	
column	decade
nameColumn	
parentColumn	
nullParentValue	
ordinalColumn	
type	Integer
internalType	
uniqueMembers	☐
levelType	TimeYears
hideMemberIf	Never
approxRowCount	
caption	
captionColumn	
formatter	
visible	☑

Level for 'Time Hierarchy' Hierarchy

Attribute	Value
name	Year
description	
table	
column	year
nameColumn	
parentColumn	
nullParentValue	
ordinalColumn	
type	Integer
internalType	
uniqueMembers	☐
levelType	TimeYears
hideMemberIf	Never
approxRowCount	
caption	
captionColumn	
formatter	
visible	☑

Long-run patterns and turning points

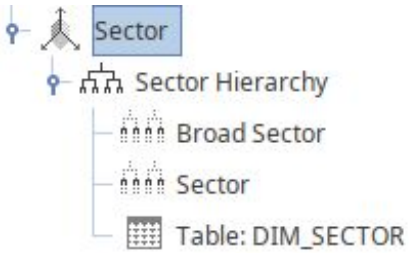
Decade summaries help identify sustained growth and convergence patterns, while annual drill-down reveals short-lived shocks that averages may conceal. Annual results also allow closer comparison with the timing of historical events

Comparable time coverage

Decade averages use available observations, so countries may be compared over different sets of years. Incomplete decades and coverage gaps must therefore be checked before interpreting differences as economic convergence or divergence.

Task 5

Sector hierarchy



Shared TimeDimension used across all three cubes

Dimension table and key

Table for 'Sector Hierarchy' Hierarchy

Attribute	Value
schema	
name	DIM_SECTOR
alias	

Hierarchy for 'Sector' Dimension

Attribute	Value
name	Sector Hierarchy
description	
hasAll	☑
allMemberName	All Sectors
allMemberCaption	
allLevelName	
defaultMember	
memberReaderClass	
primaryKeyTable	
primaryKey	sector_key
caption	
visible	☑

Hierarchy levels

Level for 'Sector Hierarchy' Hierarchy

Attribute	Value
name	Broad Sector
description	
table	
column	broad_sector
nameColumn	
parentColumn	
nullParentValue	
ordinalColumn	
type	String
internalType	
uniqueMembers	☐
levelType	Regular
hideMemberIf	Never
approxRowCount	
caption	
captionColumn	
formatter	
visible	☑

Level for 'Sector Hierarchy' Hierarchy

Attribute	Value
name	Sector
description	
table	
column	sector_name
nameColumn	
parentColumn	
nullParentValue	
ordinalColumn	
type	String
internalType	
uniqueMembers	☐
levelType	Regular
hideMemberIf	Never
approxRowCount	
caption	
captionColumn	
formatter	
visible	☑

Structural detail

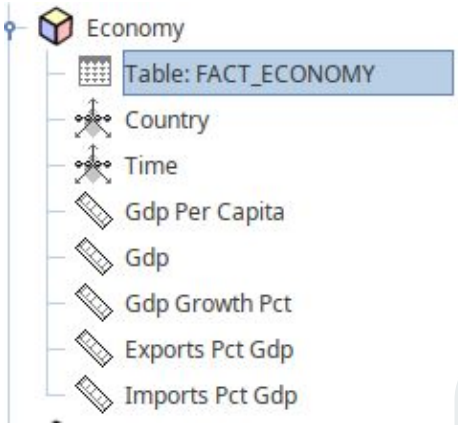
Drill-down exposes differences between industry and manufacturing that a broad-sector result can conceal. This supports a more detailed assessment of structural transformation across countries and years.

Aggregation limitation

Manufacturing is part of industry, so the two indicators overlap. With AVG, the Secondary result is their average, not a combined sector share; interpretation should therefore focus on the individual sector values

Task 5

Economy cube: dimensional integration



FACT_ECONOMY is analysed through the shared Country and Time dimensions

Dimension Usage for 'Economy' Cube

Attribute	Value
name	Country
foreignKey	country_key
source	Country
level	
usagePrefix	
caption	
visible	☑

Dimension Usage for 'Economy' Cube

Attribute	Value
name	Time
foreignKey	time_key
source	Time
level	
usagePrefix	
caption	
visible	☑

Consistent comparisons

Reusing the shared dimensions allows economic indicators to be filtered by the same country groups and periods used in the Society and Sector cubes

Complementary economic perspectives

GDP measures economic size, while GDP per capita supports comparisons between countries of different population sizes. Growth and trade indicators add information about annual performance and external exposure

Task 5

Economy cube: measures

Measure for 'Economy' Cube

Attribute	Value
name	Gdp Per Capita
description	
aggregator	avg
column	gdp_per_capita
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Economy' Cube

Attribute	Value
name	Gdp
description	
aggregator	sum
column	gdp
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Economy' Cube

Attribute	Value
name	Gdp Growth Pct
description	
aggregator	avg
column	gdp_growth_pct
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Economy' Cube

Attribute	Value
name	Exports Pct Gdp
description	
aggregator	avg
column	exports_pct_gdp
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Economy' Cube

Attribute	Value
name	Imports Pct Gdp
description	
aggregator	avg
column	imports_pct_gdp
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Average levels versus growth

AVG summarizes available GDP-per-capita levels and annual percentage indicators. A decade's average GDP growth is not cumulative decade growth, and average GDP per capita does not measure change between the decade's endpoints

Limits of aggregation

Country-group averages are unweighted and may reflect unequal coverage. GDP can be summed across non-overlapping countries for a fixed year, but a sum across years represents a multi-year total rather than an annual GDP level

Task 5

Society cube: measures



Shared Country and Time dimensions support social comparisons across countries and periods

Measure for 'Society' Cube

Attribute	Value
name	Gni Per Capita
description	
aggregator	avg
column	gni_per_capita
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Society' Cube

Attribute	Value
name	Population
description	
aggregator	sum
column	population
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Society' Cube

Attribute	Value
name	Life Expectancy
description	
aggregator	avg
column	life_expectancy
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Society' Cube

Attribute	Value
name	Urban Pop Pct
description	
aggregator	avg
column	urban_pop_pct
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Complementing GDP comparisons

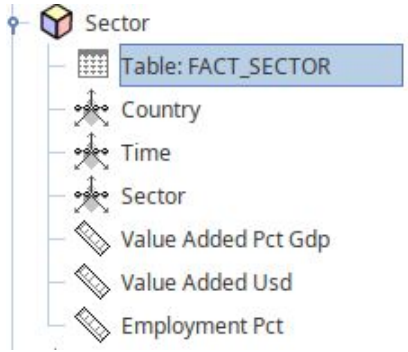
GNI per capita and life expectancy allow income and health outcomes to be compared alongside economic output. These indicators may evolve differently, so GDP rankings alone do not capture all dimensions of development.

Aggregation and comparability

AVG gives unweighted averages of available observations, not population-weighted regional indicators. Population can be summed across non-overlapping countries for a fixed year, but summing across years does not represent a population level

Task 5

Sector cube: measures



Dimension Usage for 'Sector' Cube	
Attribute	**Value**
name	Sector
foreignKey	sector_key
source	Sector
level	
usagePrefix	
caption	
visible	☑

The sector_key links fact records to the Broad Sector → Sector hierarchy

Measure for 'Sector' Cube	
Attribute	**Value**
name	Value Added Pct Gdp
description	
aggregator	avg
column	value_added_pct_gdp
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Sector' Cube	
Attribute	**Value**
name	Employment Pct
description	
aggregator	avg
column	employment_pct
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Measure for 'Sector' Cube	
Attribute	**Value**
name	Value Added Usd
description	
aggregator	sum
column	value_added_usd
formatString	
datatype	Numeric
formatter	
caption	
visible	☑

Output and employment structure

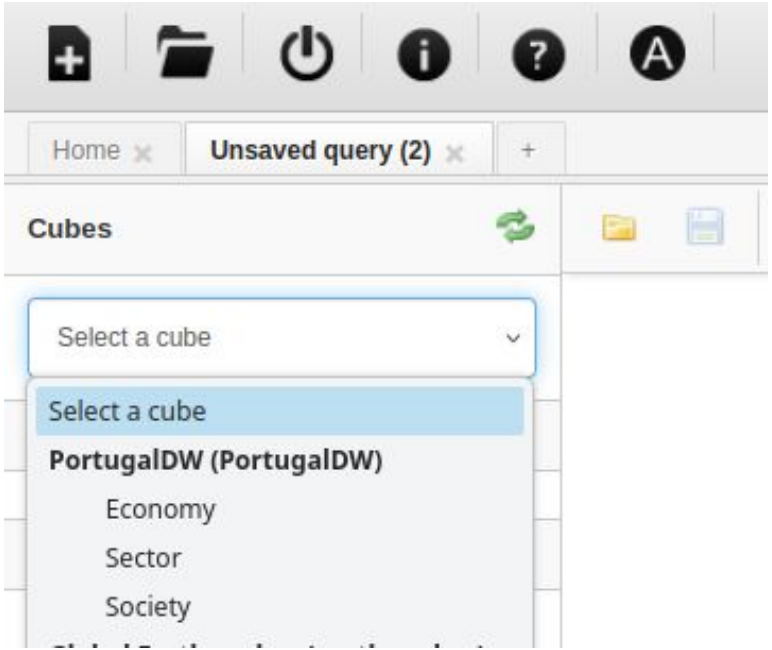
Value-added and employment shares provide complementary views of structural transformation. Comparing their evolution helps assess whether changes in a sector's contribution to GDP are accompanied by changes in its share of employment

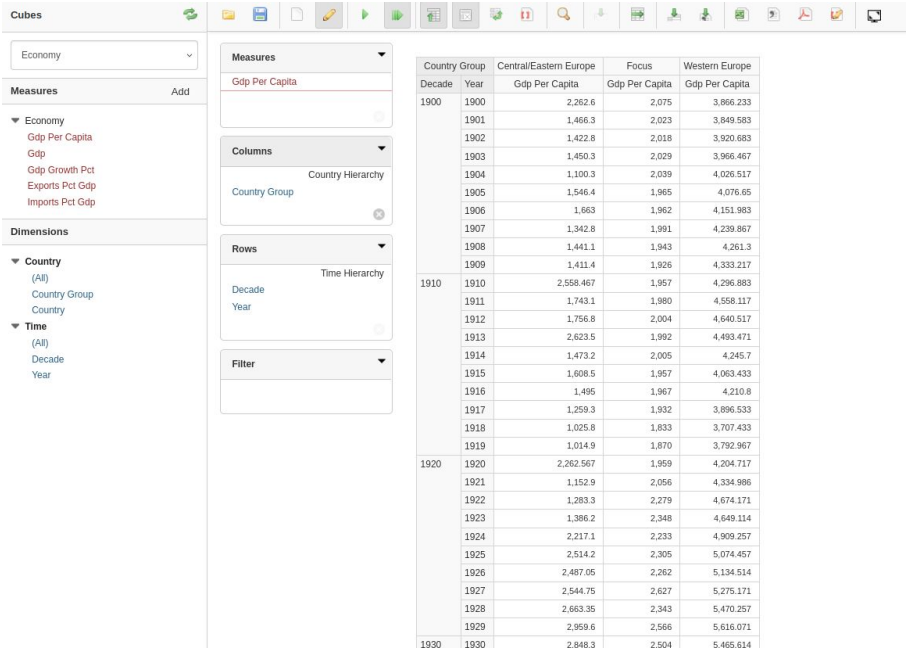
Aggregation limits

Percentage measures are averaged over available observations, rather than recomputed as weighted shares. Manufacturing overlaps with industry, so neither their USD values nor their percentages should be added as independent sector contributions

Task 5

Saiku





Country Group	Central/Eastern Europe	Focus	Western Europe
Decade	Year	Gdp Per Capita	Gdp Per Capita	Gdp Per Capita
1900	1900	2,262.6	2,075	3,866.233
	1901	1,466.3	2,023	3,849.583
	1902	1,422.8	2,018	3,920.683
	1903	1,450.3	2,029	3,966.467
	1904	1,100.3	2,039	4,026.517
	1905	1,546.4	1,965	4,076.65
	1906	1,663	1,962	4,151.983
	1907	1,342.8	1,991	4,239.867
	1908	1,441.1	1,943	4,261.3
	1909	1,411.4	1,926	4,333.217
1910	1910	2,558.467	1,957	4,296.883
	1911	1,743.1	1,980	4,558.117
	1912	1,756.8	2,004	4,640.517
	1913	2,623.5	1,992	4,493.471
	1914	1,473.2	2,005	4,245.7
	1915	1,608.5	1,957	4,063.433
	1916	1,495	1,967	4,210.8
	1917	1,259.3	1,932	3,896.533
	1918	1,025.8	1,833	3,707.433
	1919	1,014.9	1,870	3,792.967
1920	1920	2,262.567	1,959	4,204.717
	1921	1,152.9	2,056	4,334.986
	1922	1,283.3	2,279	4,674.171
	1923	1,386.2	2,348	4,649.114
	1924	2,217.1	2,233	4,909.257
	1925	2,514.2	2,305	5,074.457
	1926	2,487.05	2,262	5,134.514
	1927	2,544.75	2,627	5,275.171
	1928	2,663.35	2,343	5,470.257
	1929	2,959.6	2,566	5,616.071
1930	1930	2,848.3	2,504	5,465.614

Task 5

Main findings and limitations

- **Multidimensional integration:** three subject-specific cubes expose economic, social and sector measures. Shared Country and Time dimensions support consistent navigation across the analyses.
- **Long-run growth:** Portugal's GDP per capita increased substantially over the available period since 1900. Annual analysis reveals interruptions and reversals that decade averages can conceal.
- **Sector composition:** sector drill-down distinguishes agriculture, industry, manufacturing and services. Structural transformation requires comparisons over time; a single average across years cannot demonstrate a shift between sectors.
- **Aggregation matters:** AVG produces unweighted averages of available observations. Decade averages are not cumulative growth rates, and GDP or population summed across years should not be interpreted as annual levels.
- **Overlapping coverage:** manufacturing is included within industry, so their values cannot be added as independent contributions. Likewise, the EU aggregate must remain separate from member-country totals.
- **Uneven data availability:** coverage differs across countries, indicators and periods. Comparisons should use common observation windows, especially for Eastern Europe and incomplete decades.
- **Historical interpretation:** historical events provide context for observed turning points, but do not establish causality. Annual observations help assess whether changes coincide with the recorded events.

Task 6

Portugal vs Spain: GDP per capita

```
WITH MEMBER Measures.[Portugal vs Spain Ratio] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) /
    (Measures.[Gdp Per Capita], Country.[Country].[Spain])
SELECT
    Measures.[Portugal vs Spain Ratio] ON COLUMNS,
    {Time.Decade.[1960] : Time.Decade.[2020]} ON ROWS
FROM [Economy]
```

Decade	Portugal vs Spain Ratio
1960	0.87
1970	0.892
1980	0.912
1990	0.902
2000	0.804
2010	0.79
2020	0.839

A ratio below indicates that Portugal's GDP per capita is lower than Spain's.

Initial convergence

Portugal converged towards Spain from the 1960s to the 1980s, with the ratio increasing from 87.0% to 91.2%.

Divergence

The ratio subsequently declined to 79.0% in the 2010s, indicating divergence. The 2020s show a partial recovery to 83.9%, below the earlier peak.

Interpretation of the ratio

Portugal remained below Spain in every displayed decade. These are ratios of decade-average GDP per capita, not averages of annual ratios.

Task 6

Portugal: sector value-added shares, 1995 vs 2020

```
SELECT
    {Time.Year.[1995], Time.Year.[2020]} ON COLUMNS,
    GENERATE(
        {Sector.[Broad Sector].Members},
        DESCENDANTS(Sector.CurrentMember, Sector.[Sector], SELF_AND_BEFORE)
    ) ON ROWS
FROM [Sector]
WHERE (Measures.[Value Added Pct Gdp], Country.Country.[Portugal])
```

Broad Sector	Sector	1995	2020
Primary		4.8	2.1
	Agriculture	4.8	2.1
Secondary		20.35	15.75
	Industry	24.8	19.4
	Manufacturing	15.9	12.1
Tertiary		58.5	65.6
	Services	58.5	65.6

GENERATE and DESCENDANTS display broad groups and their sectors. The slicer fixes Portugal and Value Added Pct Gdp, while the columns compare 1995 and 2020.

Service orientation

Services increased from 58.5% to 65.6% of GDP, a rise of 7.1 percentage points.

Agriculture and industry

Agriculture declined from 4.8% to 2.1%, while Industry fell from 24.8% to 19.4%. Manufacturing decreased from 15.9% to 12.1%.

Hierarchy interpretation

Secondary displays the average of Industry and Manufacturing: 20.35% in 1995 and 15.75% in 2020. It is not a combined sector share, and Manufacturing must not be added to Industry because it is already included within it.

Task 7

A. Portugal's long-run GDP per capita

```
WITH MEMBER Measures.[Previous Decade GDP] AS
    (Measures.[Gdp Per Capita], Time.Decade.CurrentMember.PrevMember)
MEMBER Measures.[Growth Pct] AS
    IIF(
        Time.Decade.CurrentMember.PrevMember IS NULL,
        NULL,
        (Measures.[Gdp Per Capita] - Measures.[Previous Decade GDP]) / Measures.[Previous Decade GDP]
    ),
    FORMAT_STRING = '0.0%'
SELECT
    {Measures.[Gdp Per Capita], Measures.[Growth Pct]} ON COLUMNS,
    {Time.Decade.[1900] : Time.Decade.[2020]} ON ROWS
FROM [Economy]
WHERE Country.Country.[Portugal]
```

Decade	Gdp Per Capita	Growth Pct
1900	1,997.1	-
1910	1,949.7	-2.4%
1920	2,297.8	17.9%
1930	2,679	16.6%
1940	2,976.8	11.1%
1950	3,868.6	30.0%
1960	6,236.2	61.2%
1970	10,740.1	72.2%
1980	13,892.3	29.3%
1990	19,552.67	40.7%
2000	24,399.78	24.8%
2010	25,583.82	4.9%
2020	27,276.967	6.6%

Long-run evolution

Portugal's average GDP per capita rose from 1,997.1 in the 1900s to 27,277.0 in 2020–2022, despite an initial decline in the 1910s.

Strongest growth

The largest increases between consecutive decade averages were 72.2% from the 1960s to the 1970s and 61.2% from the 1950s to the 1960s.

Historical context

Industrialization and EFTA membership contextualize the acceleration. The 1973 oil shock and 1974–1975 restructuring interrupted growth, while stagnation and financial crises contextualize the weaker increase between the 2000s and 2010s.

Growth compares consecutive decade averages; it is not within-decade endpoint growth. The 2020s cover only 2020–2022.

Task 7

B. Portugal versus Western European countries

```
WITH
MEMBER Measures.[PT vs ES] AS (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Spain]), FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs FR] AS (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[France]), FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs DE] AS (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Germany]), FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs BE] AS (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Belgium]), FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs IT] AS (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Italy]), FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs IE] AS
    IIF(IsEmpty((Measures.[Gdp Per Capita], Country.[Country].[Ireland])), NULL,
        (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Ireland])),
    FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs GR] AS (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Greece]), FORMAT_STRING='0.0%'
SELECT
    {Measures.[PT vs ES], Measures.[PT vs FR], Measures.[PT vs DE], Measures.[PT vs BE],
     Measures.[PT vs IT], Measures.[PT vs IE], Measures.[PT vs GR]} ON COLUMNS,
    {Time.Decade.[1900] : Time.Decade.[2020]} ON ROWS
FROM [Economy]
```

Calculated ratios compare Portugal's decade-average GDP per capita with each country.
Values above 100% indicate a higher Portuguese level.

Task 7

B. Portugal versus Western European countries

Decade	PT vs ES	PT vs FR	PT vs DE	PT vs BE	PT vs IT	PT vs IE	PT vs GR
1900	71.9%	42.8%	40.6%	32.6%	56.5%	-	83.4%
1910	65.4%	39.0%	39.0%	32.0%	50.6%	44.7%	87.6%
1920	63.3%	36.4%	41.9%	31.2%	53.3%	54.9%	70.4%
1930	79.1%	38.9%	40.0%	34.7%	57.0%	56.9%	68.6%
1940	87.1%	52.6%	42.3%	41.6%	69.3%	60.1%	117.7%
1950	87.8%	39.8%	44.0%	39.8%	53.2%	64.2%	97.9%
1960	87.0%	43.4%	43.8%	46.7%	51.1%	77.4%	90.4%
1970	89.2%	51.6%	54.9%	54.3%	61.3%	93.4%	87.6%
1980	91.2%	55.5%	57.9%	57.6%	61.5%	93.5%	94.0%
1990	90.2%	66.2%	68.2%	66.5%	68.9%	78.7%	111.8%
2000	80.4%	69.4%	66.6%	68.0%	70.7%	52.6%	95.6%
2010	79.0%	68.6%	57.7%	65.4%	74.5%	48.8%	109.3%
2020	83.9%	72.3%	59.4%	67.3%	79.1%	47.4%	115.6%

Portugal remained below Spain, France, Germany, Belgium and Italy in every displayed decade. In 2020–2022, it reached 83.9% of Spain's level, 72.3% of France's, 59.4% of Germany's, 67.3% of Belgium's and 79.1% of Italy's.

Ireland's relative lead widened substantially: Portugal fell from 93.5% of Ireland's level in the 1980s to 47.4% in 2020–2022. Against Greece, Portugal's position fluctuated and reached 115.6% in the latest period.

Task 7

C. Convergence and Divergence

Decade	PT vs ES	PT vs FR	PT vs DE	PT vs BE	PT vs IT	PT vs IE	PT vs GR
1900	71.9%	42.8%	40.6%	32.6%	56.5%	-	83.4%
1910	65.4%	39.0%	39.0%	32.0%	50.6%	44.7%	87.6%
1920	63.3%	36.4%	41.9%	31.2%	53.3%	54.9%	70.4%
1930	79.1%	38.9%	40.0%	34.7%	57.0%	56.9%	68.6%
1940	87.1%	52.6%	42.3%	41.6%	69.3%	60.1%	117.7%
1950	87.8%	39.8%	44.0%	39.8%	53.2%	64.2%	97.9%
1960	87.0%	43.4%	43.8%	46.7%	51.1%	77.4%	90.4%
1970	89.2%	51.6%	54.9%	54.3%	61.3%	93.4%	87.6%
1980	91.2%	55.5%	57.9%	57.6%	61.5%	93.5%	94.0%
1990	90.2%	66.2%	68.2%	66.5%	68.9%	78.7%	111.8%
2000	80.4%	69.4%	66.6%	68.0%	70.7%	52.6%	95.6%
2010	79.0%	68.6%	57.7%	65.4%	74.5%	48.8%	109.3%
2020	83.9%	72.3%	59.4%	67.3%	79.1%	47.4%	115.6%

Uses the same calculated ratios as Question B to assess changes in relative position.

Broad convergence

The strongest broad catch-up occurred between the 1960s and 1990s. Portugal's ratios rose from 43.4% to 66.2% against France, 43.8% to 68.2% against Germany and 46.7% to 66.5% against Belgium.

Uneven continuation

Between the 1990s and 2000s, Portugal improved relative to France, Belgium and Italy, but lost ground against Spain, Germany and Ireland. Convergence was therefore not uniform.

Later divergence and partial recovery

Between the 2000s and 2010s, ratios declined against most comparators, with Italy and Greece as exceptions. The latest period shows partial recovery against several countries, but continued divergence from Ireland.

EFTA membership, EEC accession and monetary integration provide historical context. Decade comparisons do not isolate the effects of the 1986, 1999 or 2002 milestones.

Task 7

D.Post-transition comparison

```
WITH
MEMBER Measures.[PT vs PL] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Poland]),
    FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs CZ] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Czechia]),
    FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs HU] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Hungary]),
    FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs RO] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Romania]),
    FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs EST] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Estonia]),
    FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs LV] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Latvia]),
    FORMAT_STRING='0.0%'
MEMBER Measures.[PT vs LT] AS
    (Measures.[Gdp Per Capita], Country.[Country].[Portugal]) / (Measures.[Gdp Per Capita], Country.[Country].[Lithuania]),
    FORMAT_STRING='0.0%'
SELECT
    {Measures.[PT vs PL], Measures.[PT vs CZ], Measures.[PT vs HU], Measures.[PT vs RO],
     Measures.[PT vs EST], Measures.[PT vs LV], Measures.[PT vs LT]} ON COLUMNS,
    {Time.Decade.[1990] : Time.Decade.[2020]} ON ROWS
FROM [Economy]
```

In the 1990s, Portugal's decade-average GDP per capita exceeded that of all seven comparison countries, ranging from 136.4% of Czechia's level to 337.8% of Romania's. By 2020–2022, Portugal was below Poland, Czechia, Hungary, Estonia and Lithuania, with ratios of 89.2%, 87.0%, 97.8%, 92.1% and 90.5%. It remained slightly above Latvia and Romania, at 103.0% and 106.1%.
The narrowing gaps are consistent with post-transition recovery and European integration. EU accession in 2004 and 2007 provides context, but the ratios alone do not establish its causal contribution.

Decade	PT vs PL	PT vs CZ	PT vs HU	PT vs RO	PT vs EST	PT vs LV	PT vs LT
1990	207.2%	136.4%	189.5%	337.8%	137.2%	186.7%	193.8%
2000	155.2%	113.0%	142.0%	213.5%	115.1%	151.5%	155.3%
2010	105.3%	90.0%	111.5%	130.2%	104.4%	117.8%	107.7%
2020	89.2%	87.0%	97.8%	106.1%	92.1%	103.0%	90.5%

The 1990 row represents the entire 1990s, including transition and recovery years; it is not the position immediately after communism ended.

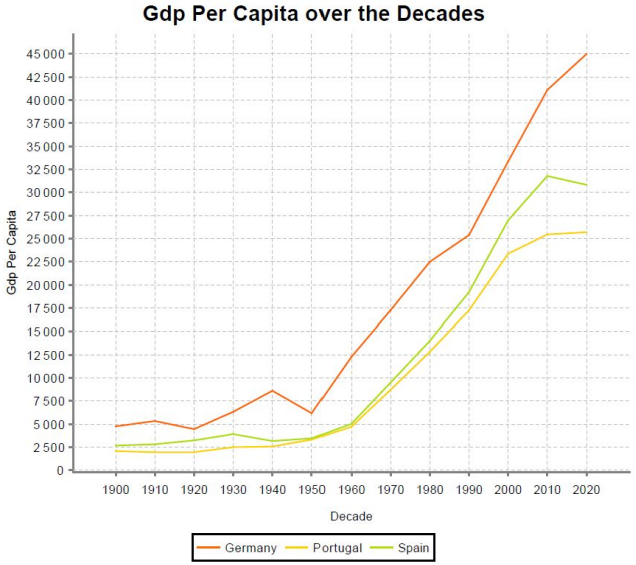
Task 8

Part A - PRD report

```sql
SELECT dt.Year, e.event_description, e.category, e.economic_impact, dc.country_name
FROM portugal_dw.EVENTS e
JOIN portugal_dw.DIM_COUNTRY dc ON e.country_key  = dc.country_key
JOIN portugal_dw.DIM_TIME dt ON e.time_key  = dt.time_key
WHERE dc.iso3_code in ('PRT', 'DEU', 'ESP')
ORDER BY dt.Year;
```

```sql
SELECT dc.country_name,
       dc.country_group,
       dt.year,
       fe.gdp_per_capita,
       fs.gni_per_capita,
       COALESCE(fs.life_expectancy, 'NA')
FROM portugal_dw.FACT_ECONOMY fe
JOIN portugal_dw.DIM_COUNTRY dc ON dc.country_key = fe.country_key
JOIN portugal_dw.DIM_TIME dt ON fe.time_key = dt.time_key
LEFT JOIN portugal_dw.FACT_SOCIETY fs ON fe.country_key = fs.country_key AND fe.time_key = fs.time_key
WHERE dc.country_name IN ('Portugal', 'Spain', 'Germany')
AND dt.year % 10 = 0
ORDER BY dc.country_name desc, dt.year
```

Portugal's Economic and Human Development: A Century of Transformation



Key events: Germany's early lead (1900-1945) reflects its status as Europe's industrial powerhouse, interrupted by two world wars and hyperinflation (1923), before the postwar Wirtschaftswunder (1950s) rebuilds its advantage. Portugal and Spain, by contrast, remain close together and far behind until the 1960s both overwhelmingly agrarian economies under authoritarian regimes (Estado Novo; Franco dictatorship). Spain's 1959 Stabilization Plan and Portugal's EFTA accession (1960) trigger parallel "economic boom" decades. Simultaneous 1986 EEC accession marks the steepest sustained convergence for both Iberian economies, narrowing the gap with Germany through the 1990s. All three diverge again after the 2008 financial crisis, most visibly Spain, whose housing bubble collapse (2008) and bank bailout (2012) produce a sharper dip than Portugal's own 2011 Troika bailout.

Task 8

Part A - PRD report

Portugal's Economic and Human Development: A Century of Transformation

Germany

Country	Life Expectancy	Year	GDP per Capita
Germany	NA	1900	4,758
Germany	NA	1910	5,337
Germany	NA	1920	4,457
Germany	NA	1930	6,333
Germany	NA	1940	8,612
Germany	NA	1950	6,186
Germany	69.1	1960	12,282
Germany	70.4	1970	17,277
Germany	72.8	1980	22,497
Germany	75.1	1990	25,391
Germany	77.9	2000	33,367.3
Germany	80	2010	41,109.6
Germany	81	2020	44,996.4

Average GDP per capita: 17,892.56

Portugal

Country	Life Expectancy	Year	GDP per Capita
Portugal	NA	1900	2,075
Portugal	NA	1910	1,957
Portugal	NA	1920	1,959
Portugal	NA	1930	2,504
Portugal	NA	1940	2,574
Portugal	NA	1950	3,325
Portugal	64	1960	4,712
Portugal	67.1	1970	8,724

Task 8

Part A - PRD report

Portugal's Economic and Human Development: A Century of Transformation

Portugal	71.2	1980	12,822
Portugal	74	1990	17,256
Portugal	76.3	2000	23,372
Portugal	79	2010	25,463.2
Portugal	81.3	2020	25,700.3

Average GDP per capita: 10,187.96

Spain

Country	Life Expectancy	Year	GDP per Capita
Spain	NA	1900	2,672.4
Spain	NA	1910	2,819.3
Spain	NA	1920	3,239.3
Spain	NA	1930	3,918.3
Spain	NA	1940	3,167

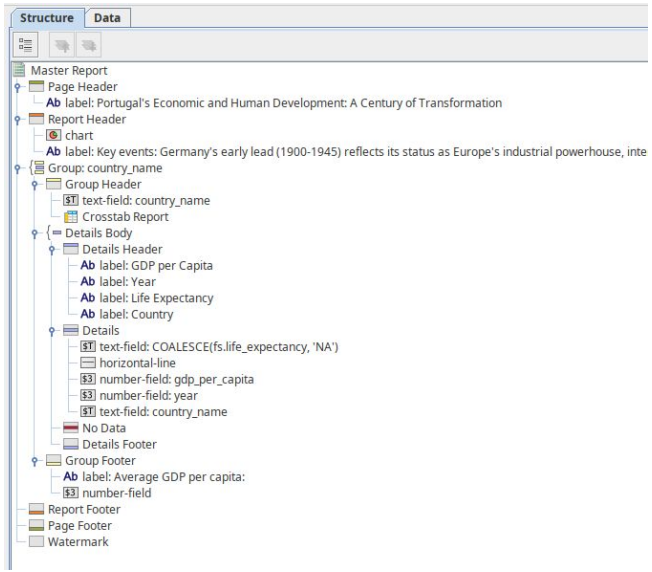
Portugal's Economic and Human Development: A Century of Transformation

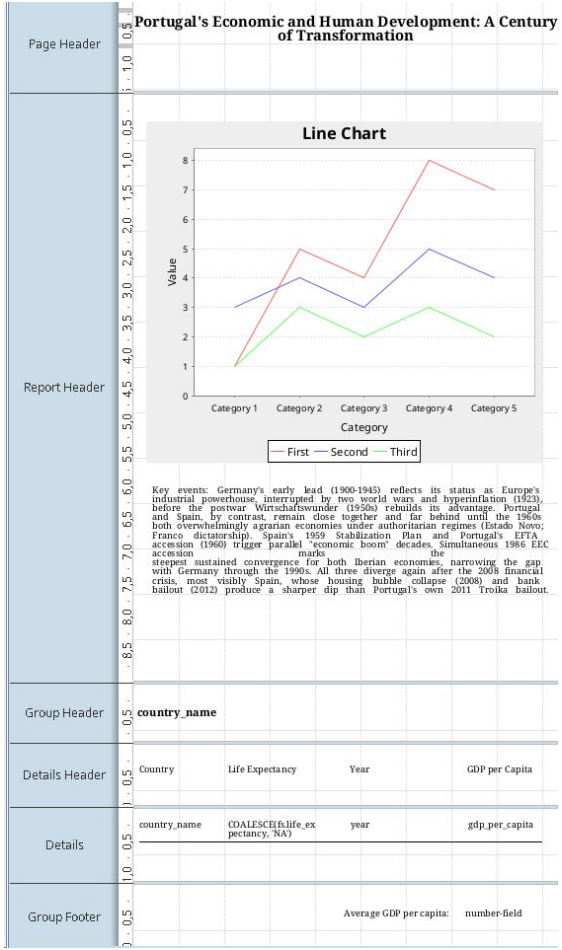
Spain	NA	1950	3,459.4
Spain	69.1	1960	5,031.6
Spain	72	1970	9,502.9
Spain	75.3	1980	14,005.9
Spain	76.8	1990	19,215
Spain	79	2000	26,994.9
Spain	81.6	2010	31,786.4
Spain	82.2	2020	30,834.5

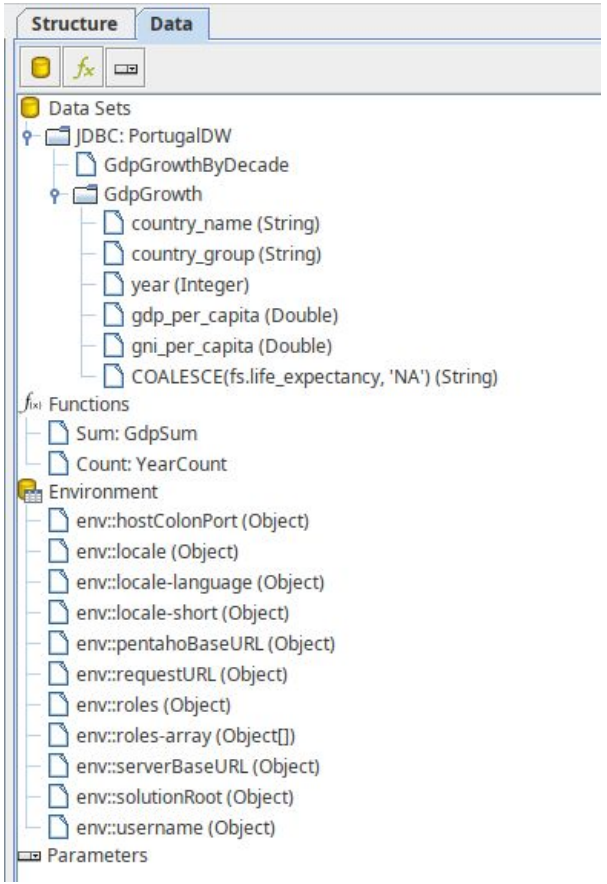
Average GDP per capita: 12,049.76

Task 8

Part A - PRD report

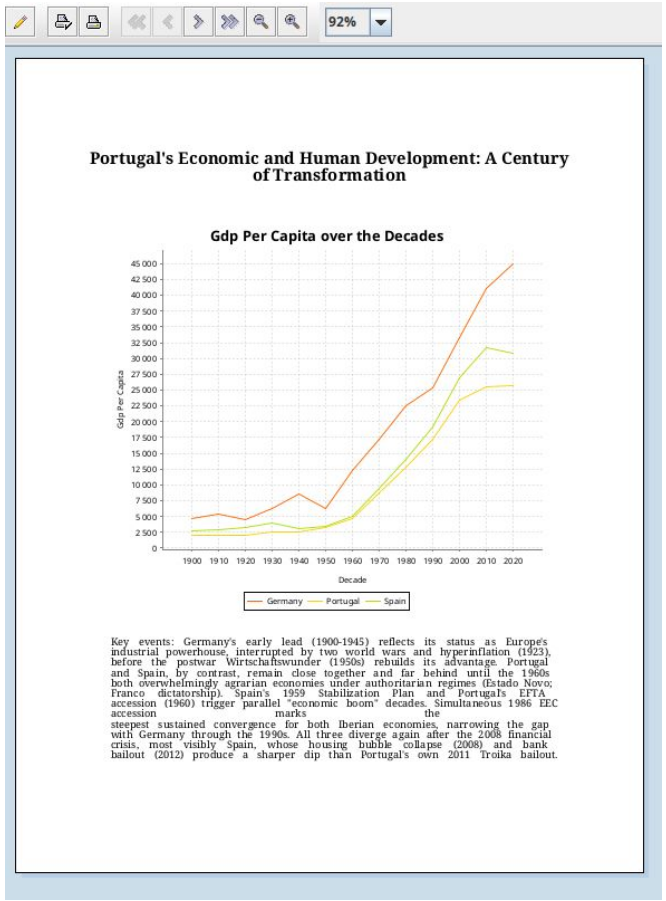






Task 8

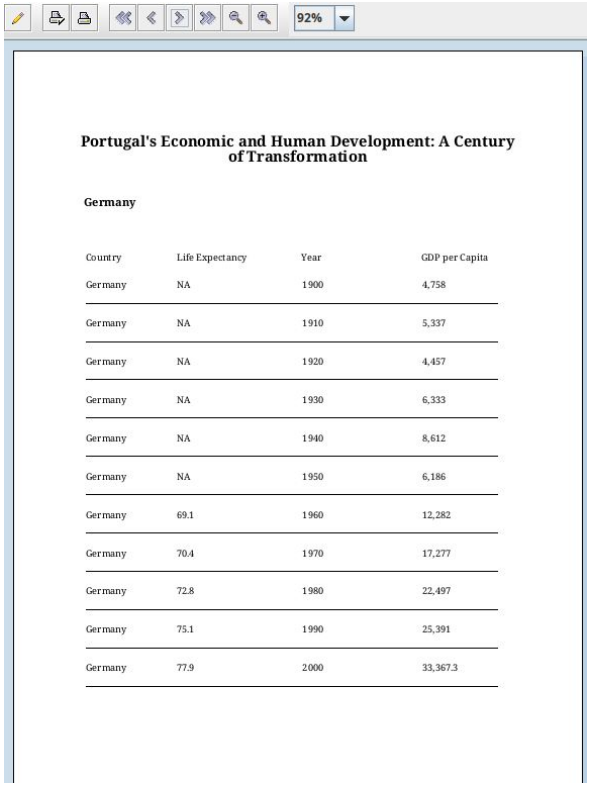
Part A - PRD report



Portugal's Economic and Human Development: A Century of Transformation

Gdp Per Capita over the Decades

Key events: Germany's early lead (1900-1945) reflects its status as Europe's industrial powerhouse, interrupted by two world wars and hyperinflation (1923), before the postwar Wirtschaftswunder (1950s) rebuilds its advantage. Portugal and Spain, by contrast, remain close together and far behind until the 1960s both overwhelmingly agrarian economies under authoritarian regimes (Estado Novo; Franco dictatorship). Spain's 1959 Stabilization Plan and Portugal's EFTA accession (1960) trigger parallel "economic boom" decades. Simultaneous 1986 EEC accession marks the steepest sustained convergence for both Iberian economies, narrowing the gap with Germany through the 1990s. All three diverge again after the 2008 financial crisis, most visibly Spain, whose housing bubble collapse (2008) and bank bailout (2012) produce a sharper dip than Portugal's own 2011 Troika bailout.



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Germany

Country	Life Expectancy	Year	GDP per Capita
Germany	NA	1900	4,758
Germany	NA	1910	5,337
Germany	NA	1920	4,457
Germany	NA	1930	6,333
Germany	NA	1940	8,612
Germany	NA	1950	6,186
Germany	69.1	1960	12,282
Germany	70.4	1970	17,277
Germany	72.8	1980	22,497
Germany	75.1	1990	25,391
Germany	77.9	2000	33,367.3

Task 8

Part A - Report findings and limitations

Long-run economic progress

Portugal's GDP per capita increased from 2,075 in 1900 to 25,700 in 2020. Despite substantial progress, it remained below Spain and Germany at every displayed comparison year.

Human development

Portugal's life expectancy increased from 64.0 years in 1960 to 81.3 in 2020. By 2020, it was close to Spain's 82.2 years and slightly above Germany's 81.0, despite Portugal's lower GDP per capita.

Sampling and summary statistics

The report displays individual years at ten-year intervals, not decade averages. Country summaries average only the displayed observations, while annual fluctuations remain hidden.

Historical context and missing data

Historical annotations contextualize the trajectories without establishing causality. NA identifies unavailable life-expectancy observations and must not be interpreted as zero.

Task 8

Parte B - Data coverage and missingness

Country reconciliation

Country names were checked across sources, while ISO3 codes provided the common integration key. The main geographical difference is the EU aggregate, which appears in WDI and events but not in the supplied Maddison data.

Different temporal windows

Maddison spans 1900–2022, whereas WDI coverage varies by indicator and country. Employment series begin later, and Portugal's sector value-added shares start in 1995, limiting comparisons over a common period.

Country-specific gaps

Late starts and internal gaps affect several historical country series. Comparisons must check actual observations, since matching minimum and maximum years does not guarantee complete coverage.

Empty EU measures

The EU aggregate has no Maddison-sourced GDP, GDP per capita or population in the warehouse. These NULLs reflect source coverage and prevent its use as a benchmark in long-run Maddison comparisons.

Task 8

Parte B - ETL bottlenecks and analytical implications

WDI long-to-wide pivot

WDI indicators had to be mapped and pivoted into country–year and country–year–sector records. Missing indicator combinations were retained as NULL, preserving the distinction between absent data and genuine zero values.

Repeatable loading

Insert / update allowed individual pipelines to be rerun while preserving existing dimension keys. This avoided rebuilding dependent tables, but did not automatically handle source deletions.

Connectivity and cube access

Container-to-MySQL connectivity and conflicting Mondrian classes initially blocked access to the cubes. Resolving these issues was necessary before the analytical results could be produced.

Reporting consequences

Uneven series coverage complicated chart construction. Selecting aligned observation years improved readability, but reduced temporal detail; displaying NA clarified missingness without resolving it.

Task 8

Parte B - Interesting patterns discovered

Uneven convergence with Spain

Portugal approached Spain between the 1960s and 1980s, but subsequently lost relative ground. The partial recovery in 2020–2022 remained below the earlier peak.

Eastern European catch-up

Portugal's relative GDP-per-capita advantage narrowed markedly against several Eastern European countries. The decade-based results show that several had overtaken Portugal in the available 2020s observations.

Shift towards services

From 1995 to 2020, services rose from 58.5% to 65.6% of GDP, while agriculture fell from 4.8% to 2.1%. Manufacturing is included within industry and must not be counted as an independent additional contribution.

Portugal's Economic and Human Development: A Century of Transformation

Data Analysis and Integration

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1st semester, 2026/2027